

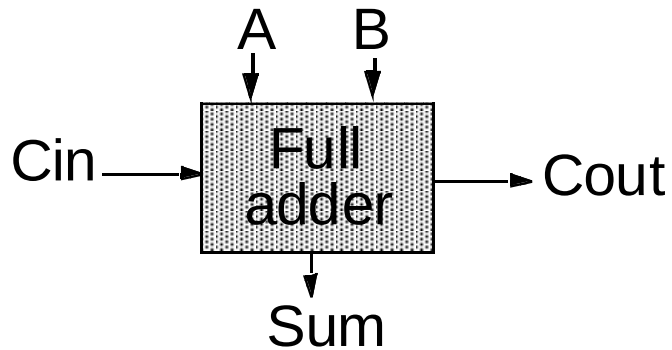
Materiale integrative per il corso di Fondamenti di Elettronica Digitale, prof. P.Palestri

varie figure prese da

<http://icbook.eecs.berkeley.edu/resources/powerpoint-slides>

"Adapted from Digital Integrated Circuits (2nd Edition): Jan M. Rabaey,
Copyright 2003 Prentice Hall/Pearson."

Adders (1)



A	B	C_i	S	C_o	<i>Carry status</i>
0	0	0	0	0	delete
0	0	1	1	0	delete
0	1	0	1	0	propagate
0	1	1	0	1	propagate
1	0	0	1	0	propagate
1	0	1	0	1	propagate
1	1	0	0	1	generate
1	1	1	1	1	generate

$$S = A \oplus B \oplus C_i$$

$$= A\bar{B}\bar{C}_i + \bar{A}B\bar{C}_i + \bar{A}\bar{B}C_i + ABC_i$$

$$C_o = AB + BC_i + AC_i$$

OR

$$\text{Generate (G)} = AB$$

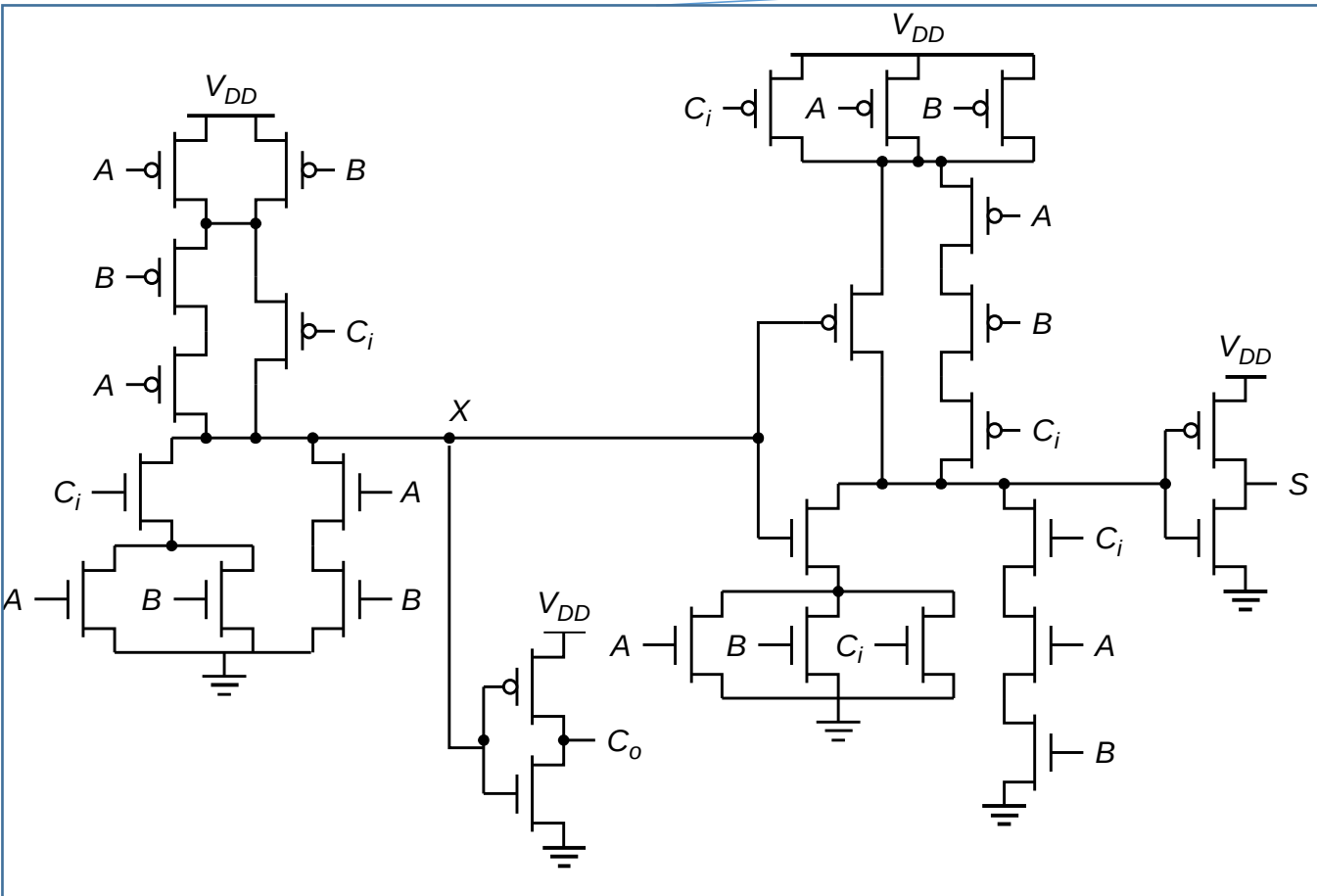
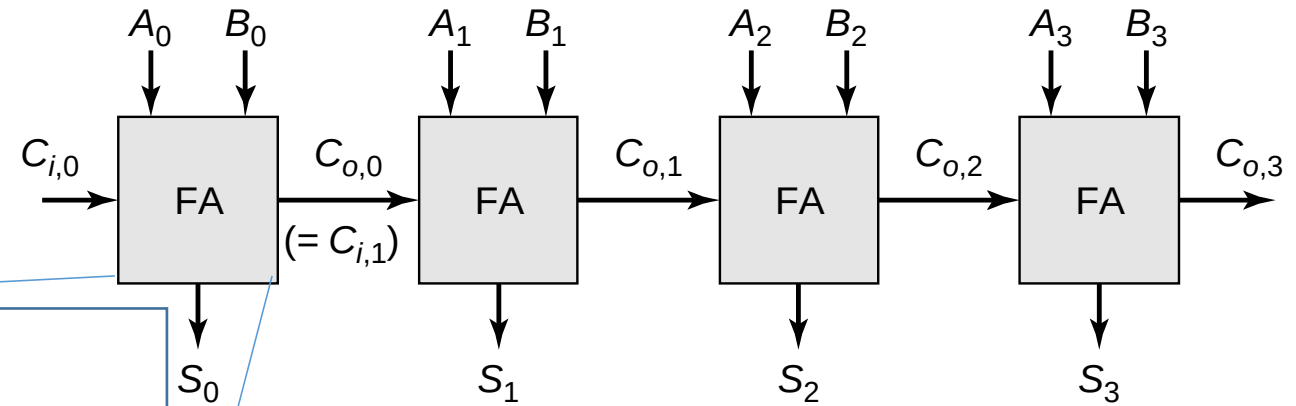
$$\text{Propagate (P)} = A \oplus B$$

$$C_o(G, P) = G + PC_i$$

$$S(G, P) = P \oplus C_i$$

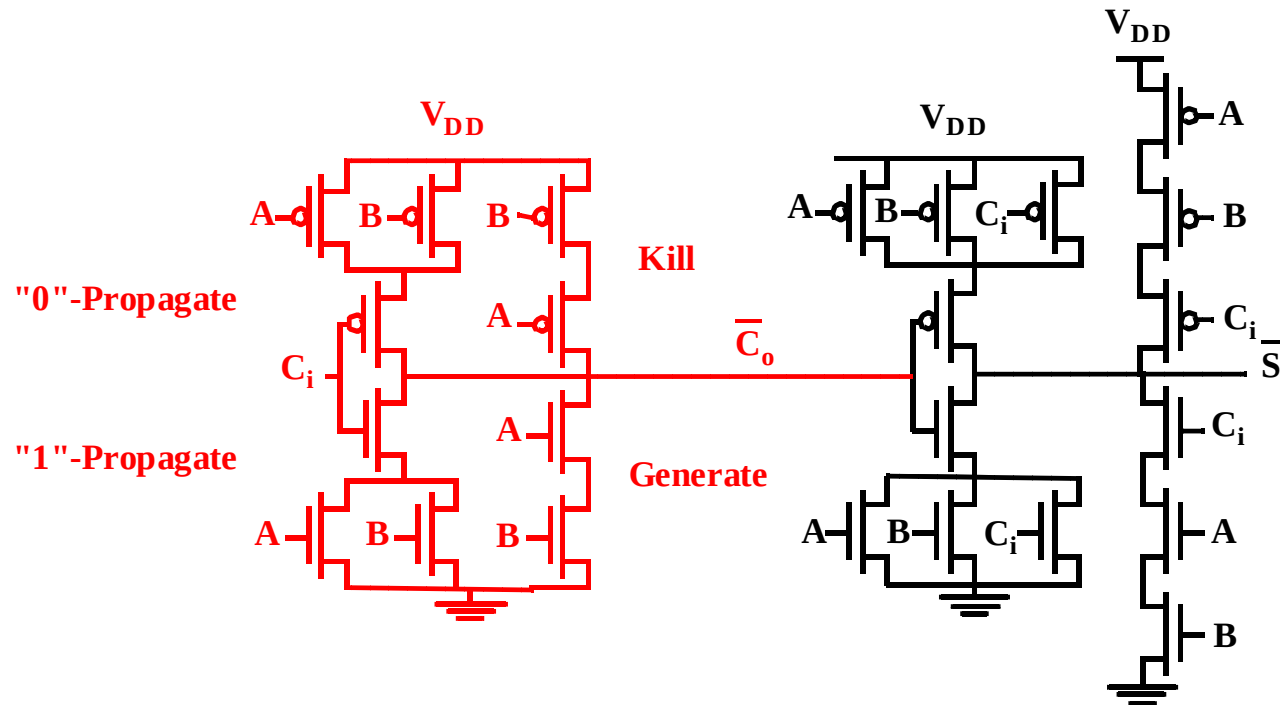
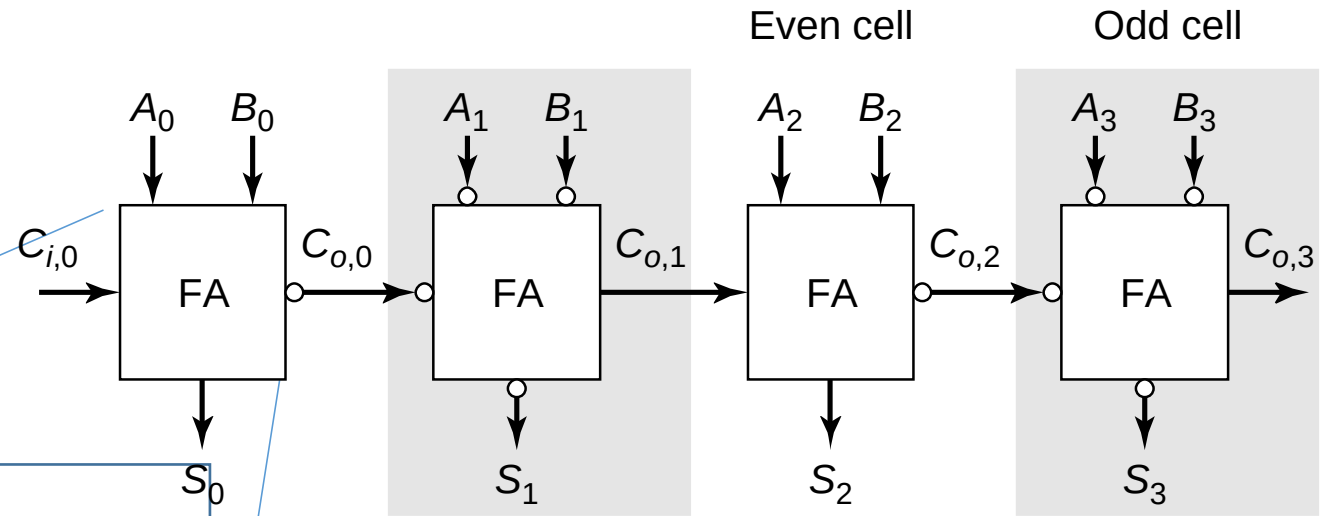
Adders (2)

carry ripple adder



$$\begin{aligned}
 S &= A \oplus B \oplus C_i \\
 &= A\bar{B}\bar{C}_i + \bar{A}B\bar{C}_i + \bar{A}\bar{B}C_i + ABC_i \\
 C_o &= AB + BC_i + AC_i
 \end{aligned}$$

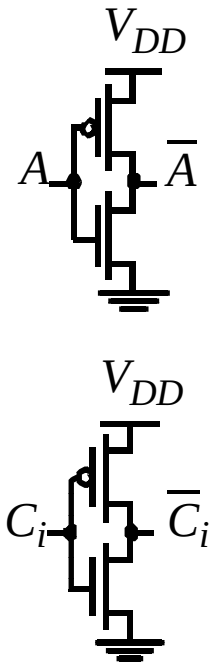
Adders (3)



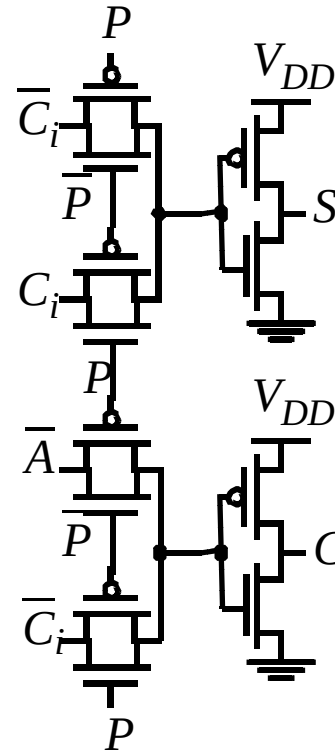
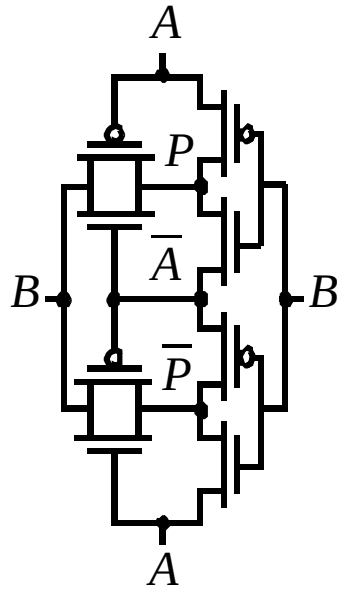
$$\begin{aligned}
 S &= A \oplus B \oplus C_i \\
 &= A\bar{B}\bar{C}_i + \bar{A}B\bar{C}_i + \bar{A}\bar{B}C_i + ABC_i \\
 C_o &= AB + BC_i + AC_i
 \end{aligned}$$

Adders (4)

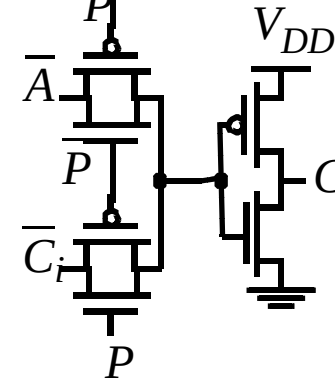
pass-transistors



Setup



Sum Generation



Carry Generation

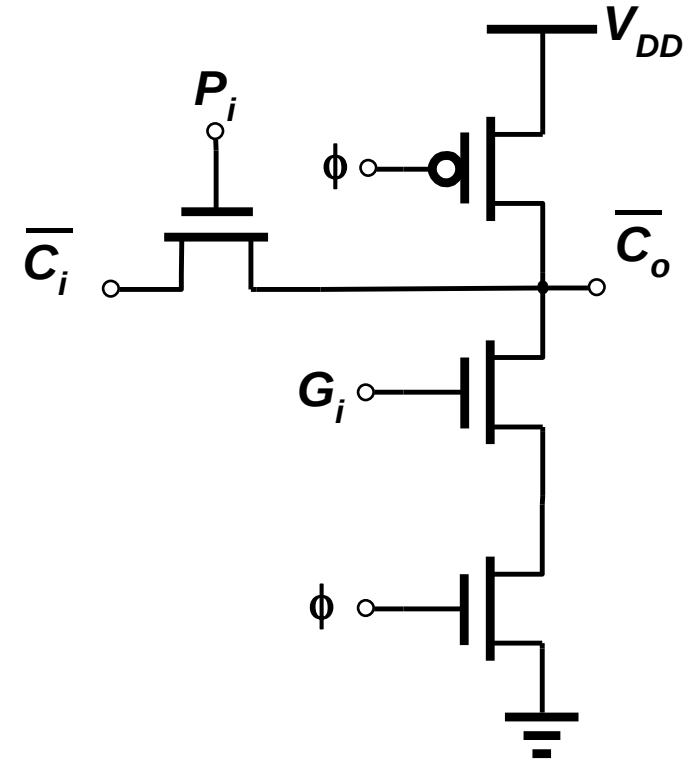
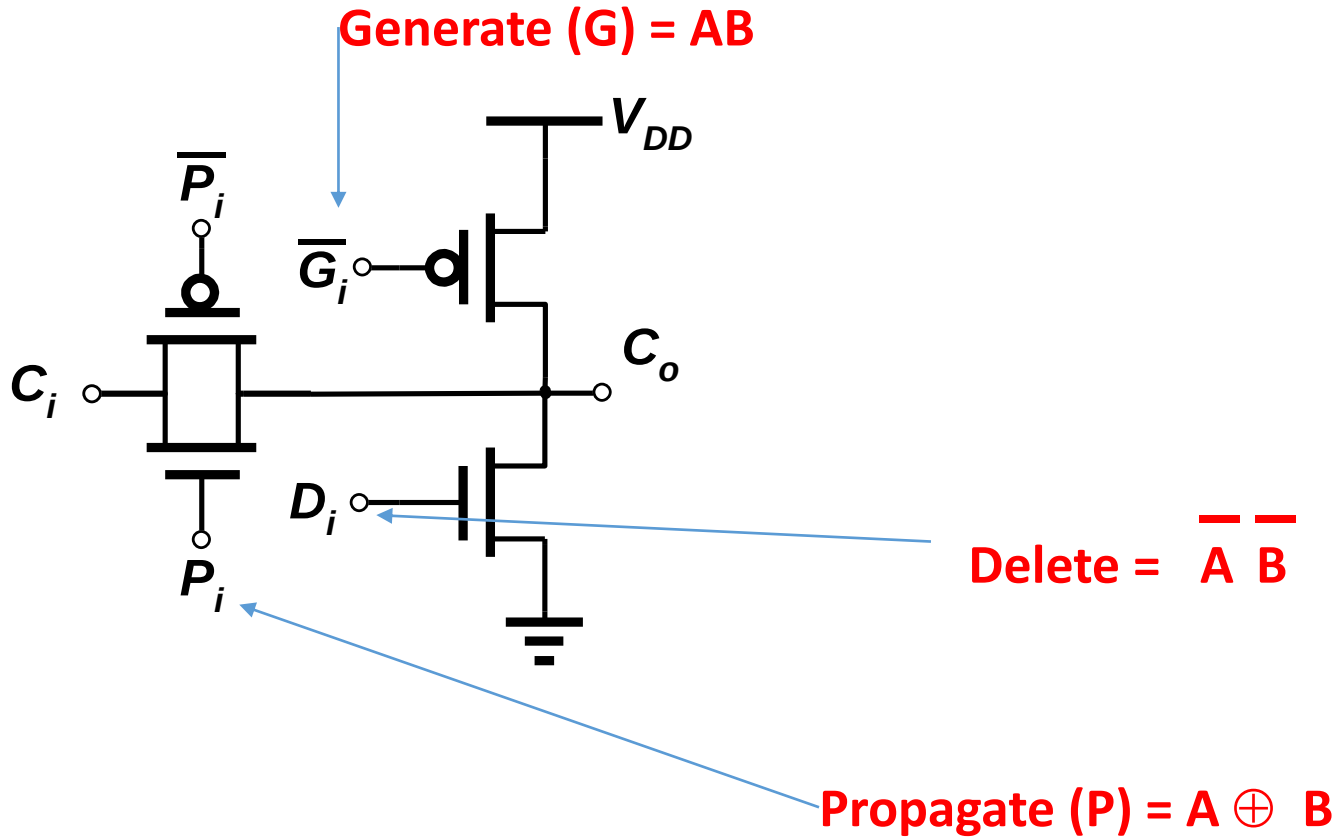
$$\text{Propagate (P)} = A \oplus B$$

$$C_o(G, P) = G + PC_i$$

$$S(G, P) = P \oplus C_i$$

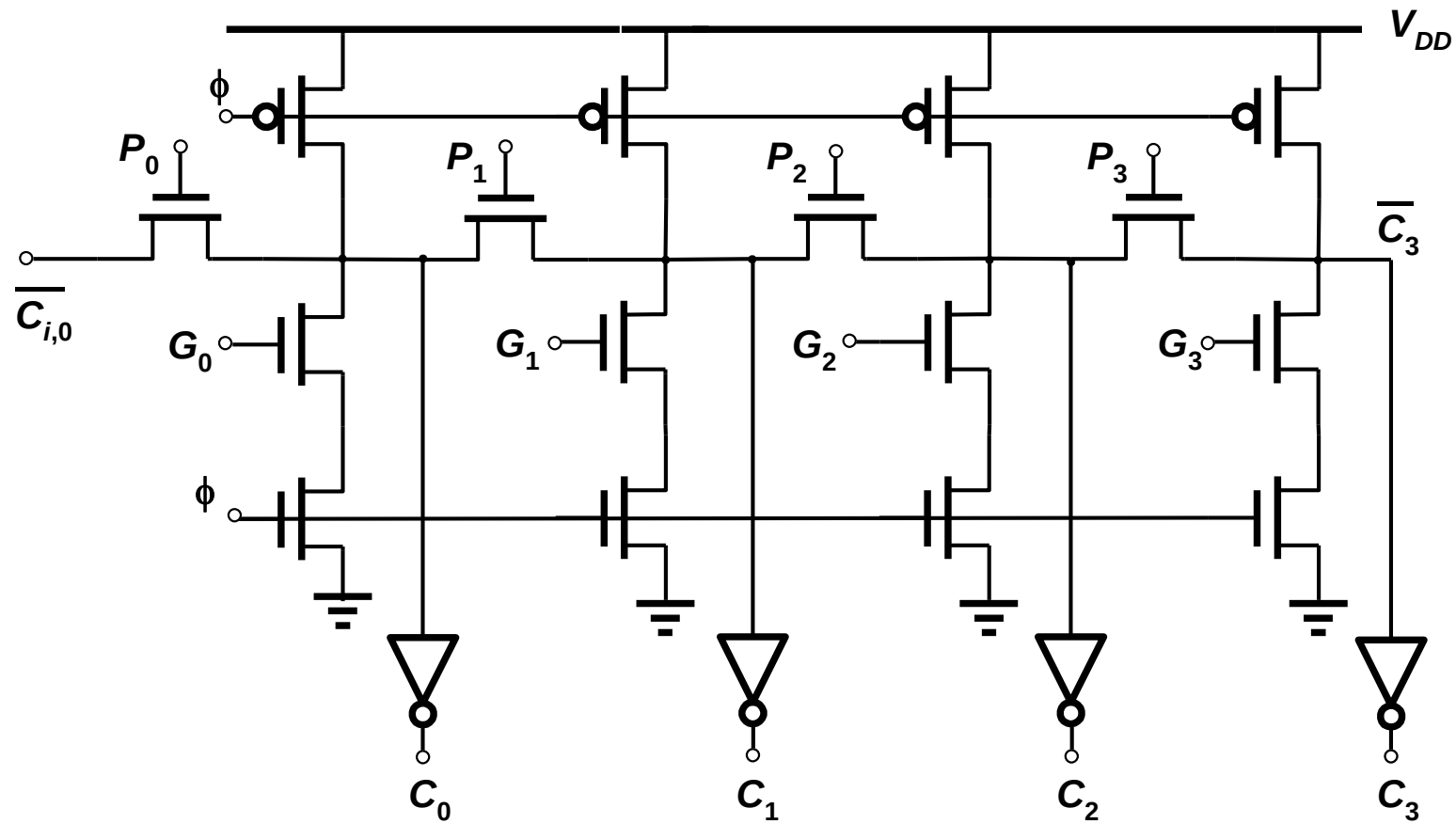
Adders (5)

$$C_o(G, P) = G + PC_i$$



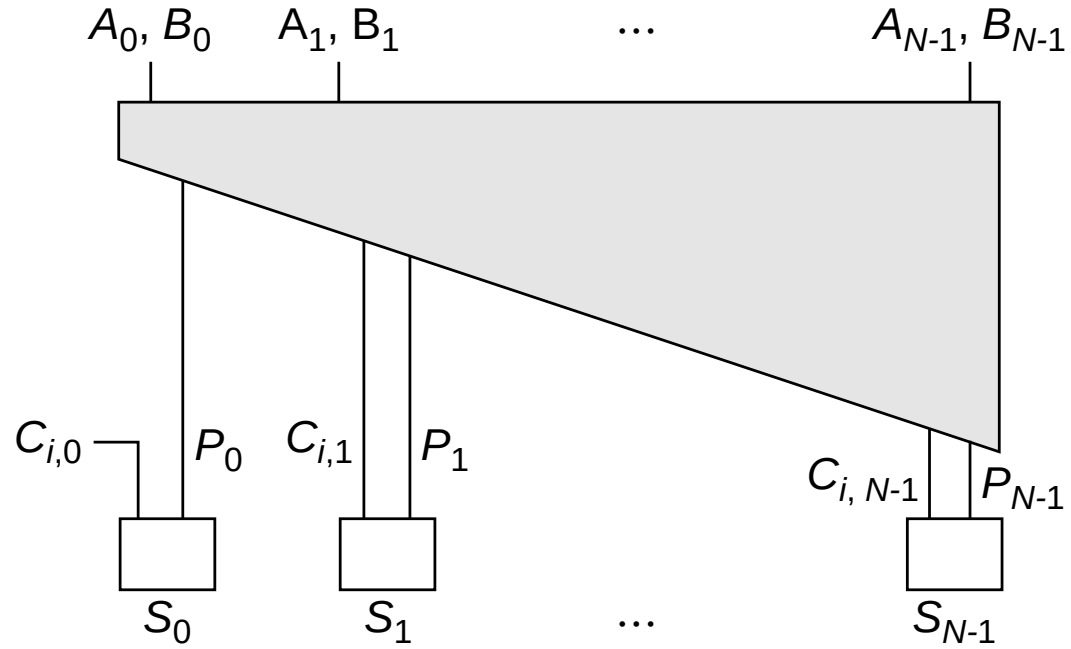
Adders (6)

Manchester carry chain

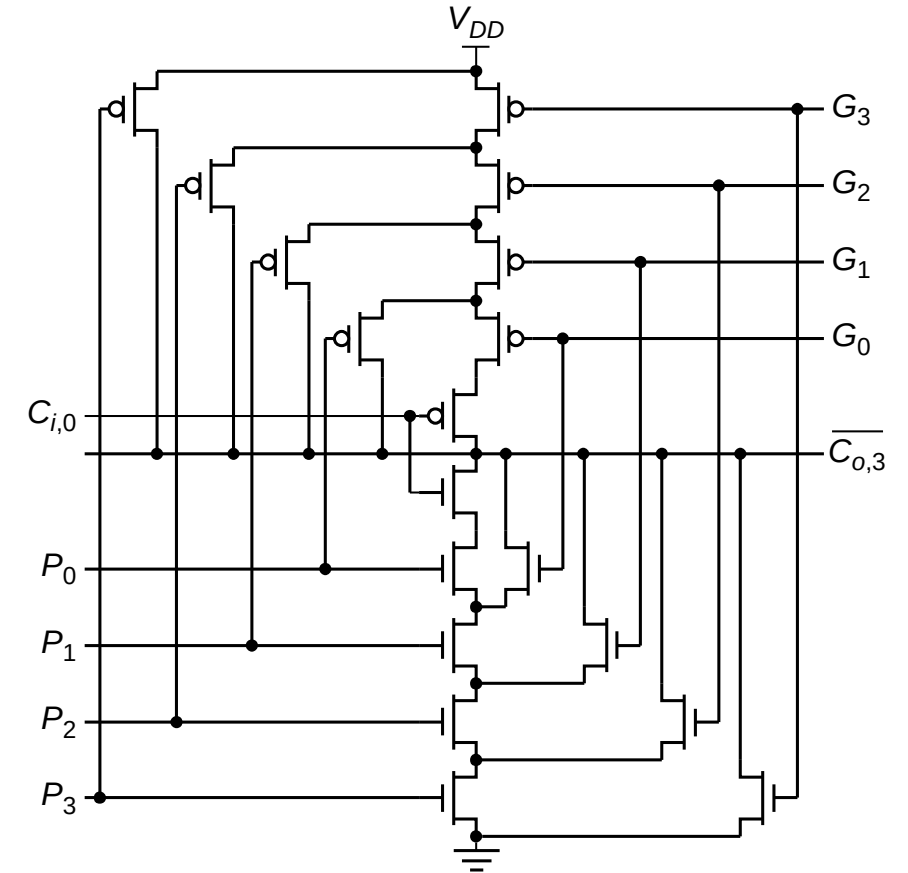


Adders (7)

- carry lookAhead

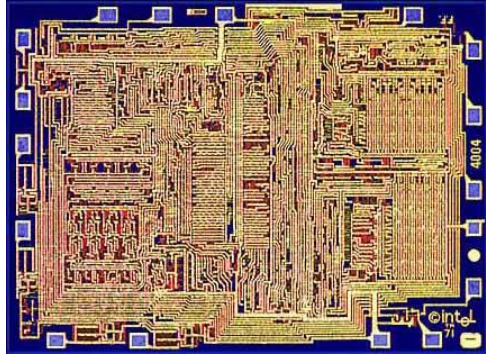


$$C_{o,k} = f(A_k, B_k, C_{o,k-1}) = G_k + P_k C_{o,k-1}$$



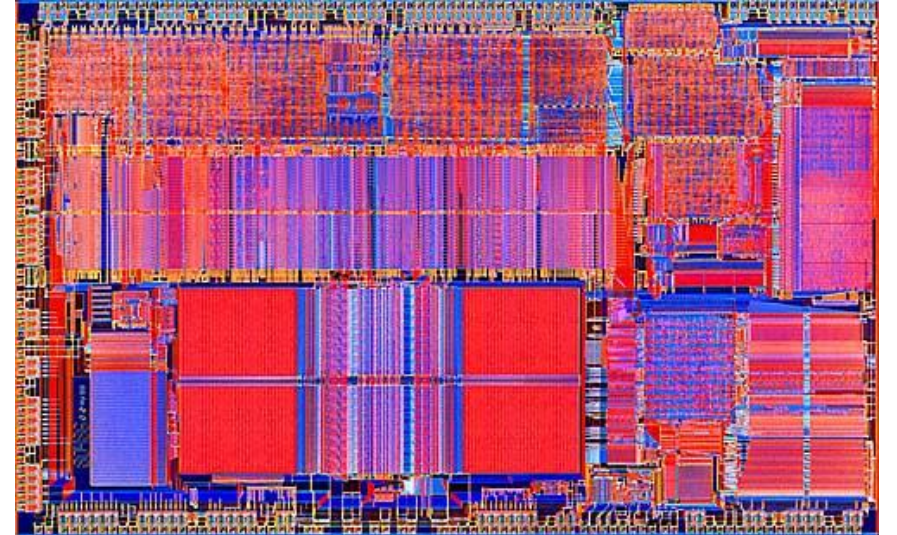
$$C_{o,k} = G_k + P_k (G_{k-1} + P_{k-1} (\dots + P_1 (G_0 + P_0 C_{i,0})))$$

Design styles: full custom



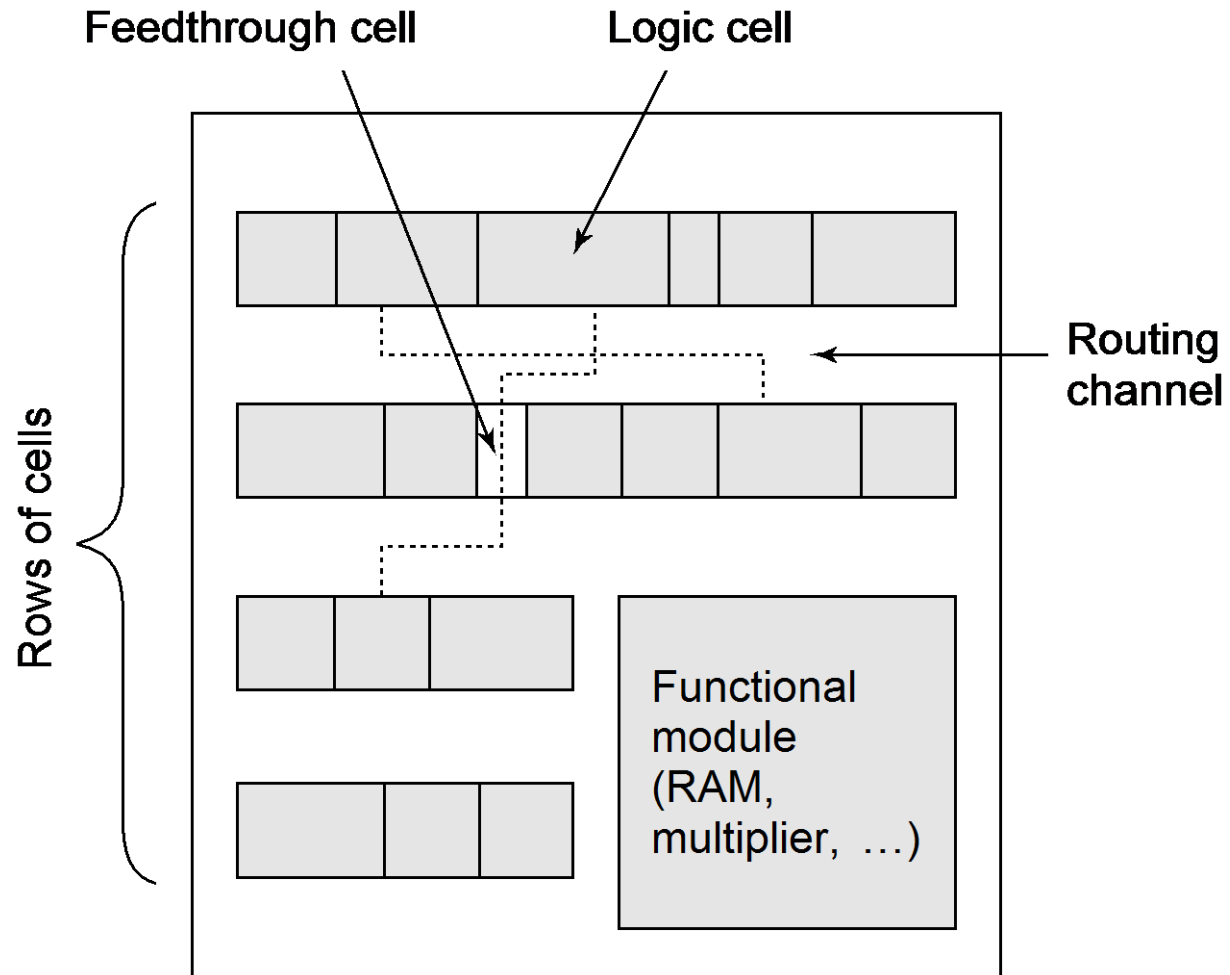
Intel 4004 ('71)

Transition to Automation and
Regular Structures

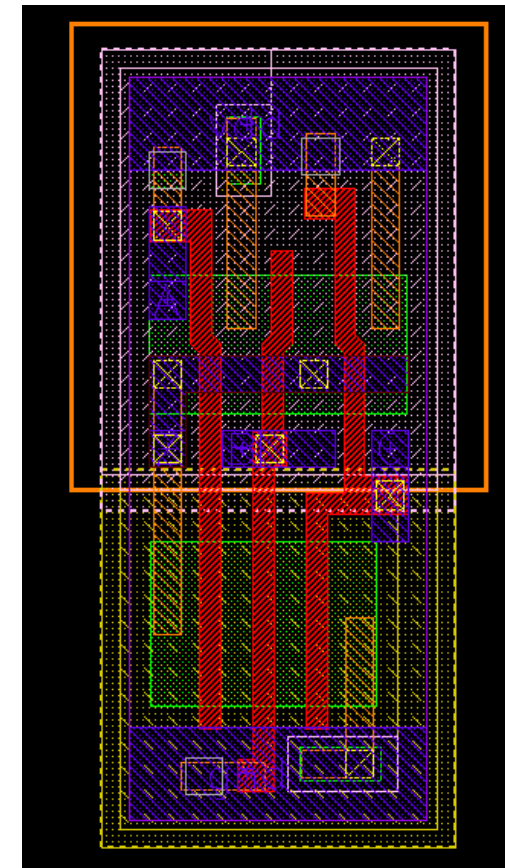


Intel 8486

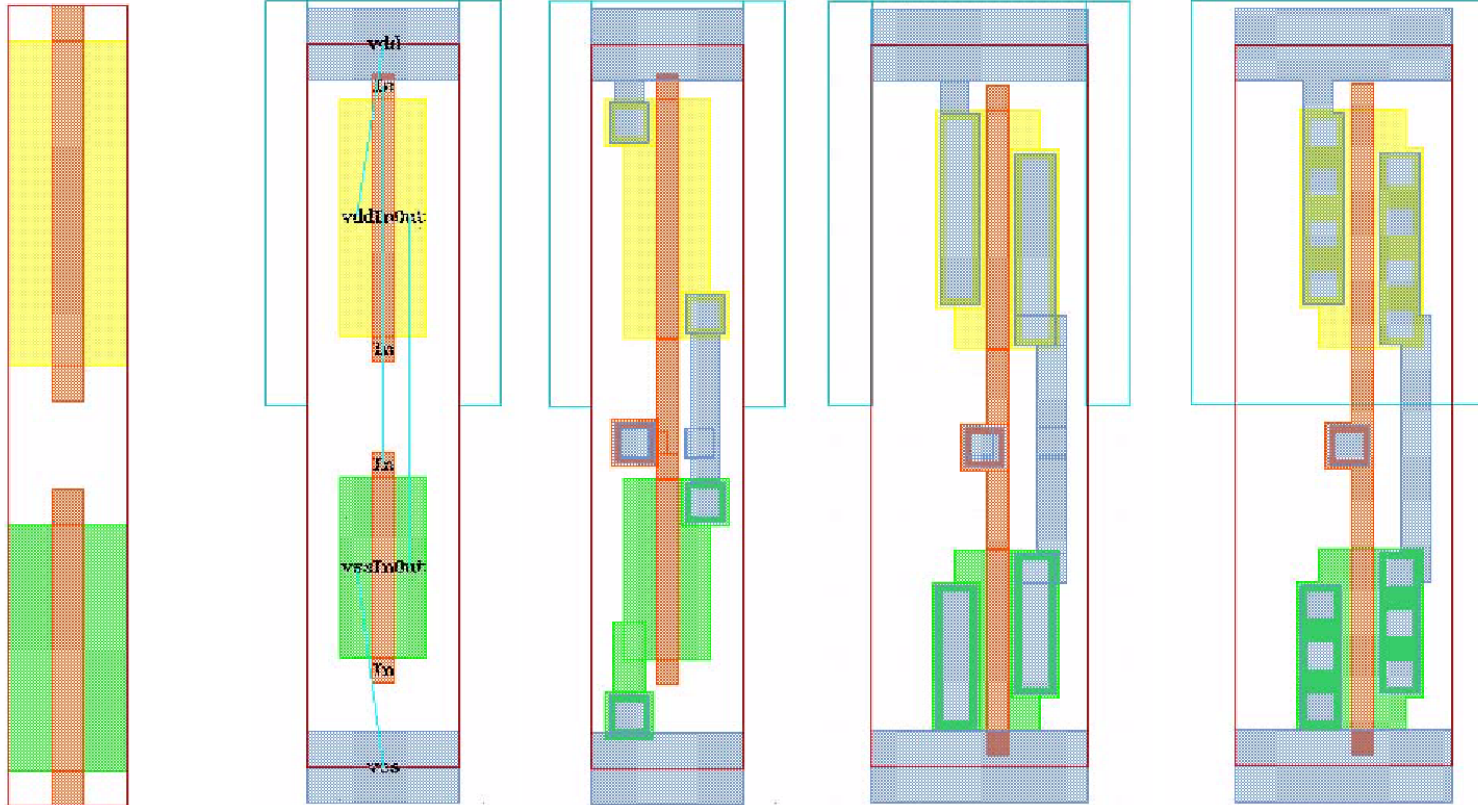
Design styles: standard cells (1)



example of cell: 3-input NAND cell
(from ST Microelectronics)



Design styles: standard cells (2)



Initial transistor geometries

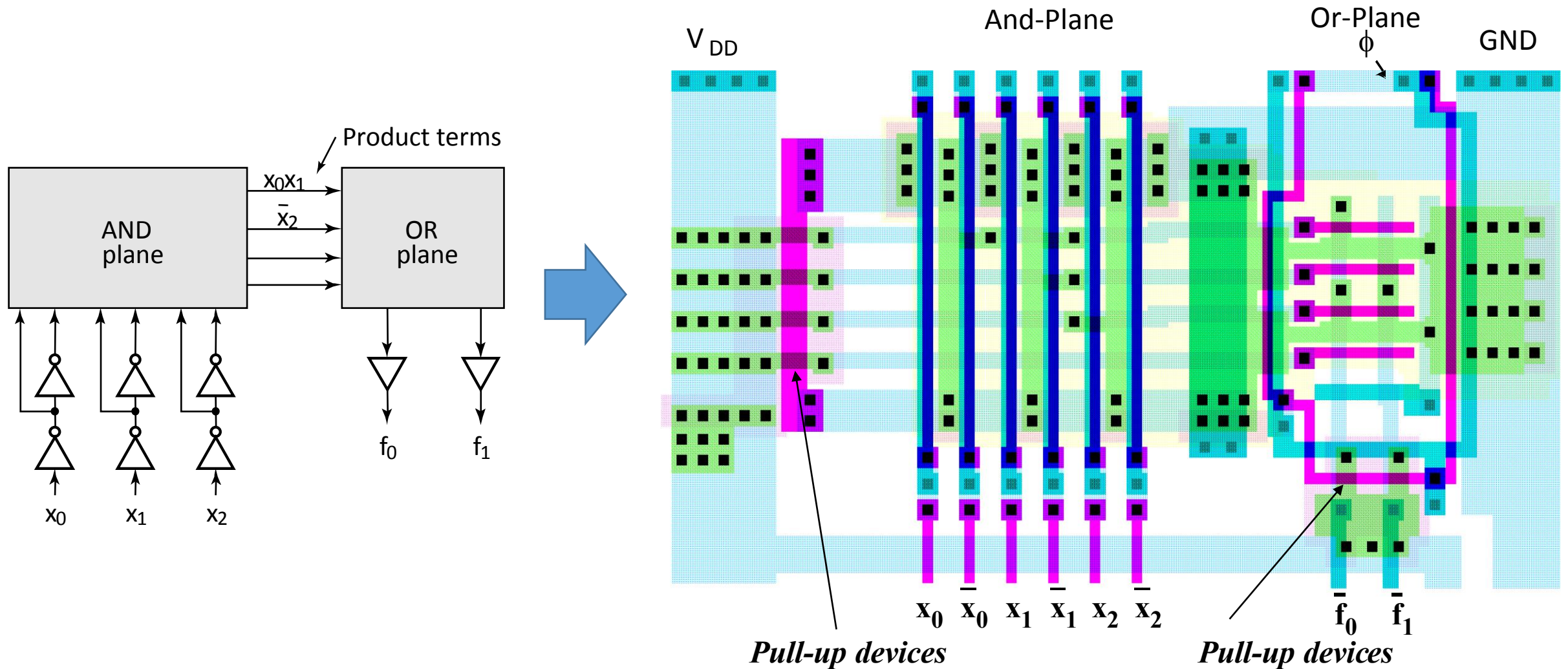
Placed transistors

Routed cell

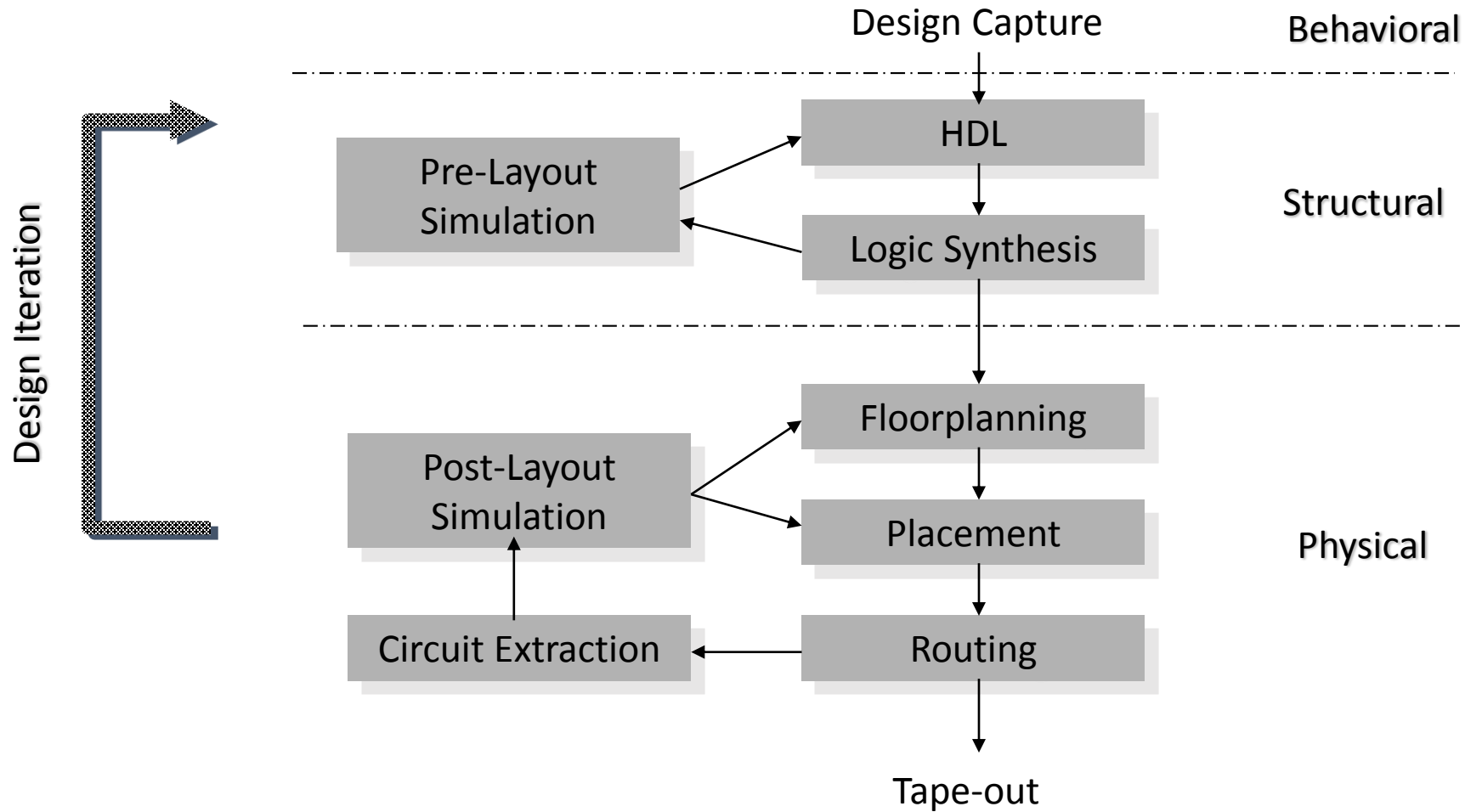
Compacted cell

Finished cell

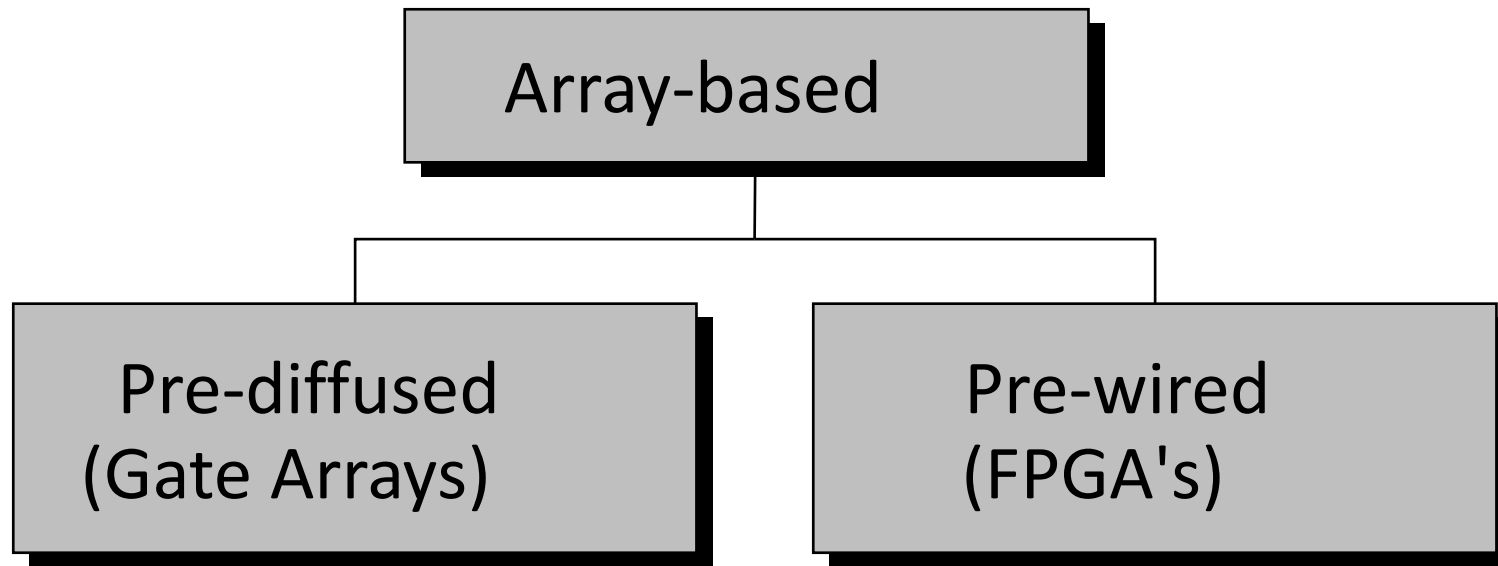
Design styles: PLA (programmable-logic-array)



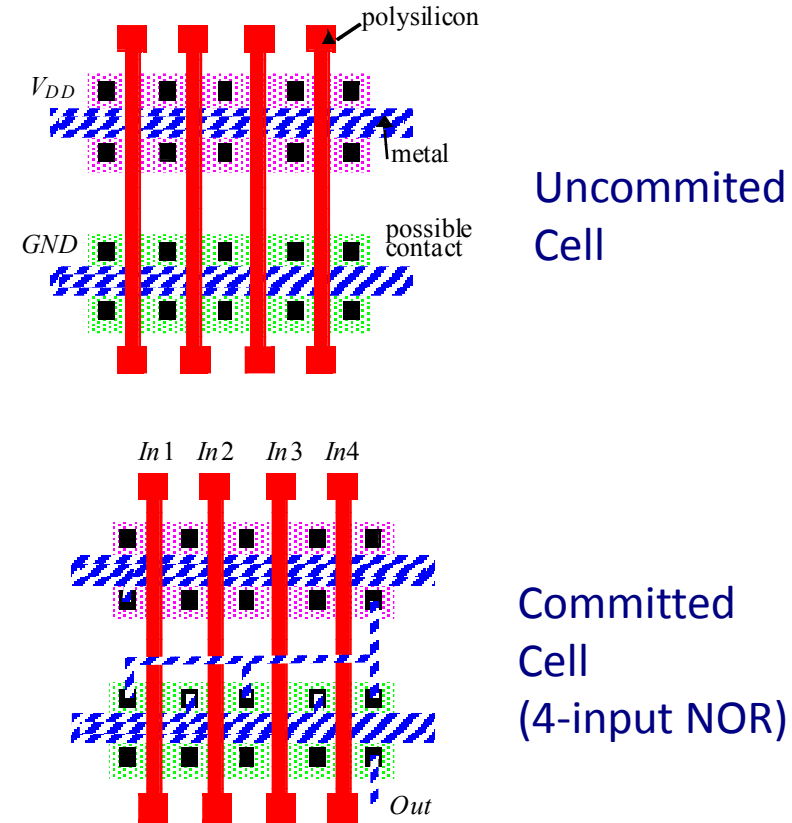
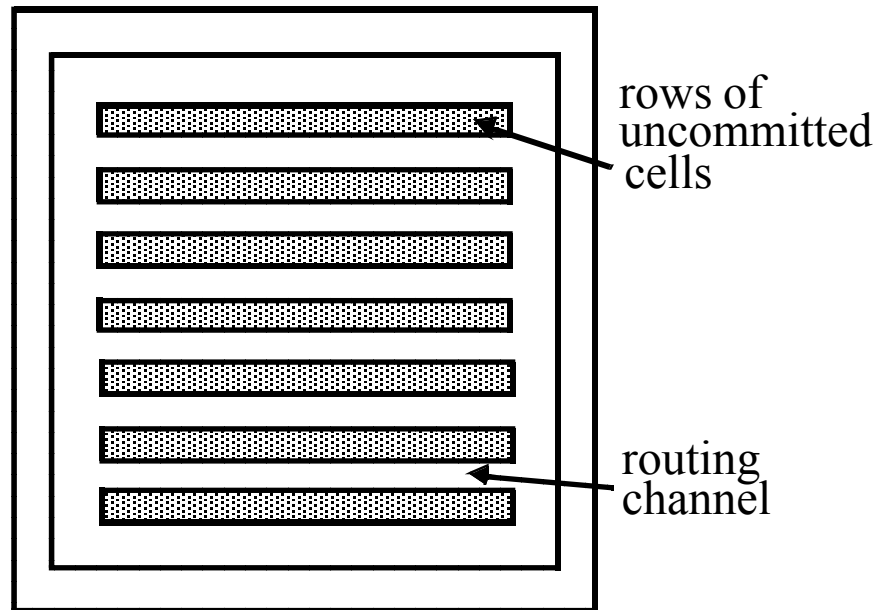
Design styles: semi-custom



Design styles: late-binding

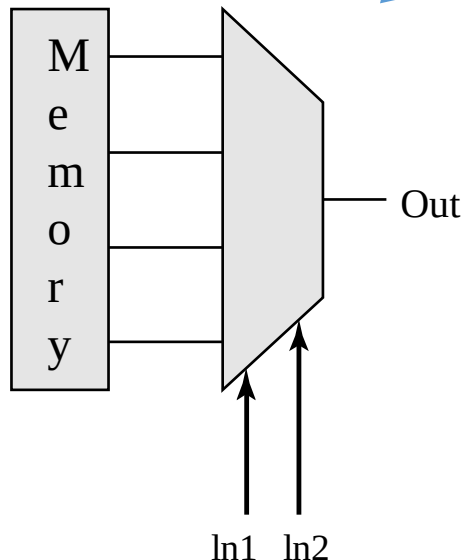


Design styles: gate-array / sea-of-gates

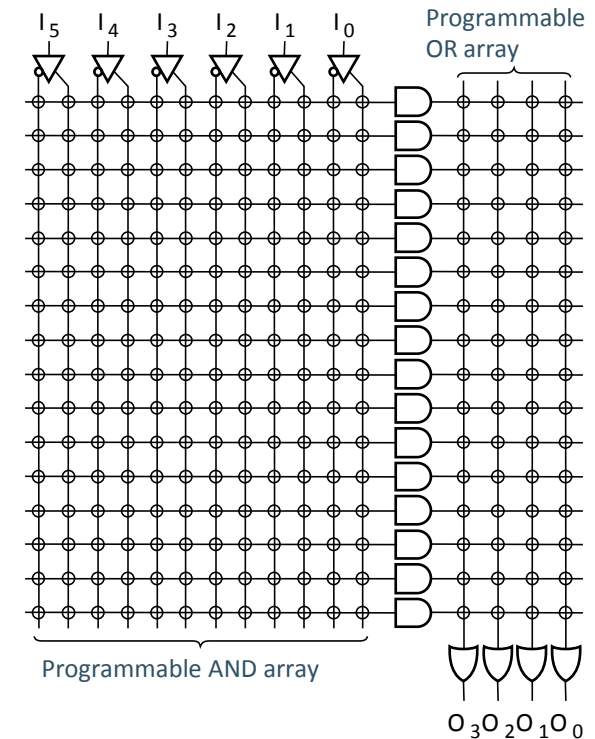


Design styles: pre-wired arrays (1)

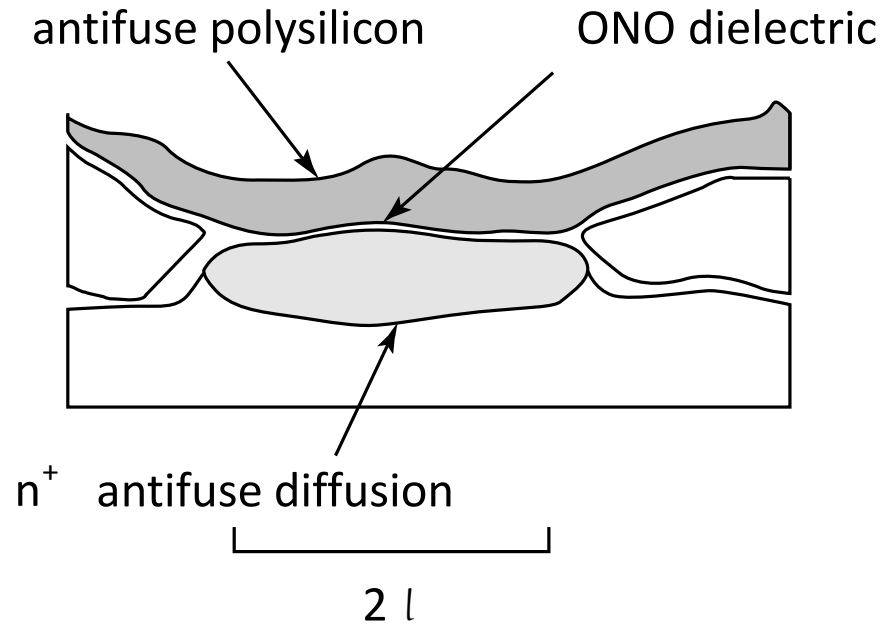
- Programming Technique
 - Fuse-based (program-once)
 - Non-volatile EPROM based
 - RAM based
- Programmable Logic Style
 - Array-Based
 - Look-up Table



In	Out
00	0
01	1
10	1
11	0



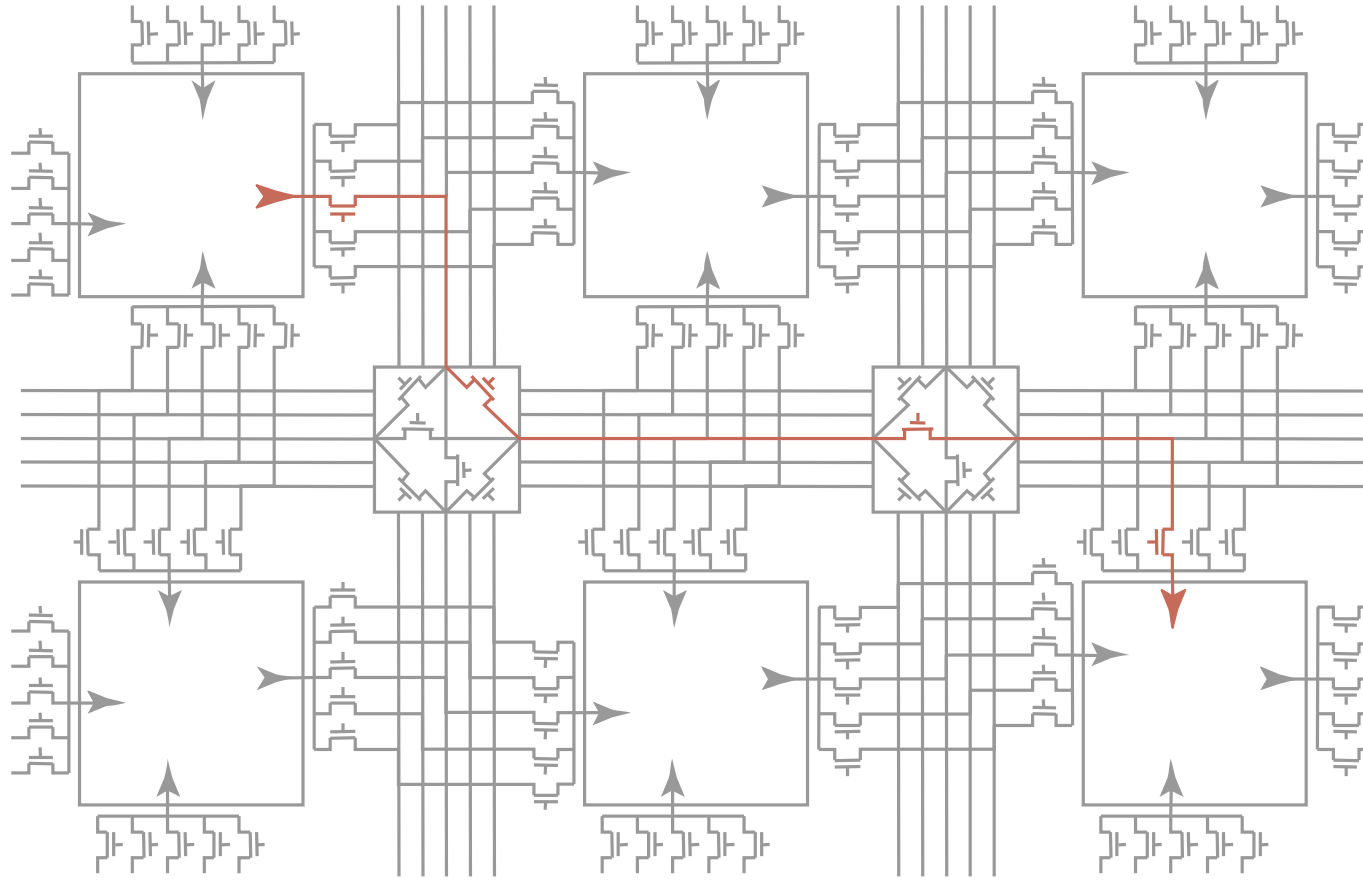
Design styles: pre-wired arrays (2)



Open by default, closed by applying current pulse

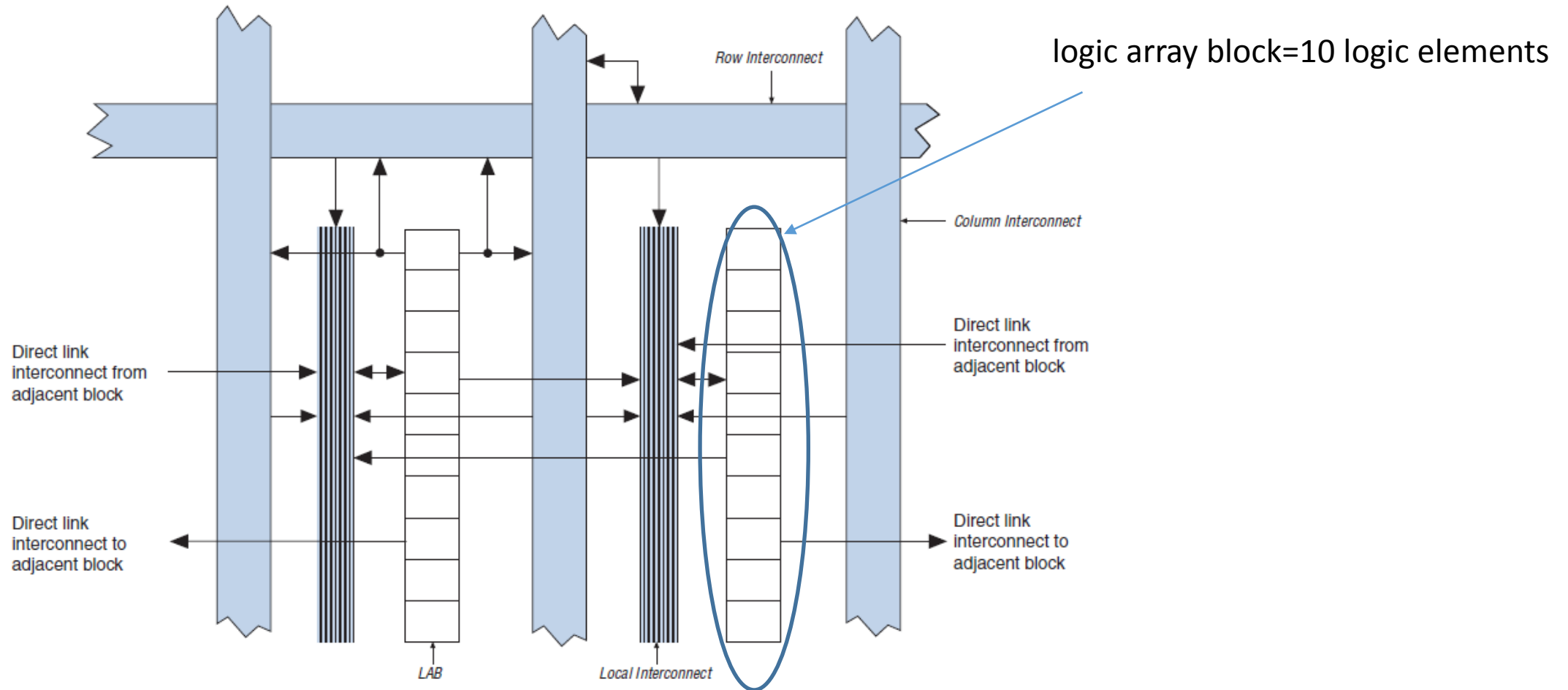
One Time Programmable (OTP)

Design styles: Mesh interconnection



FPGA (1)

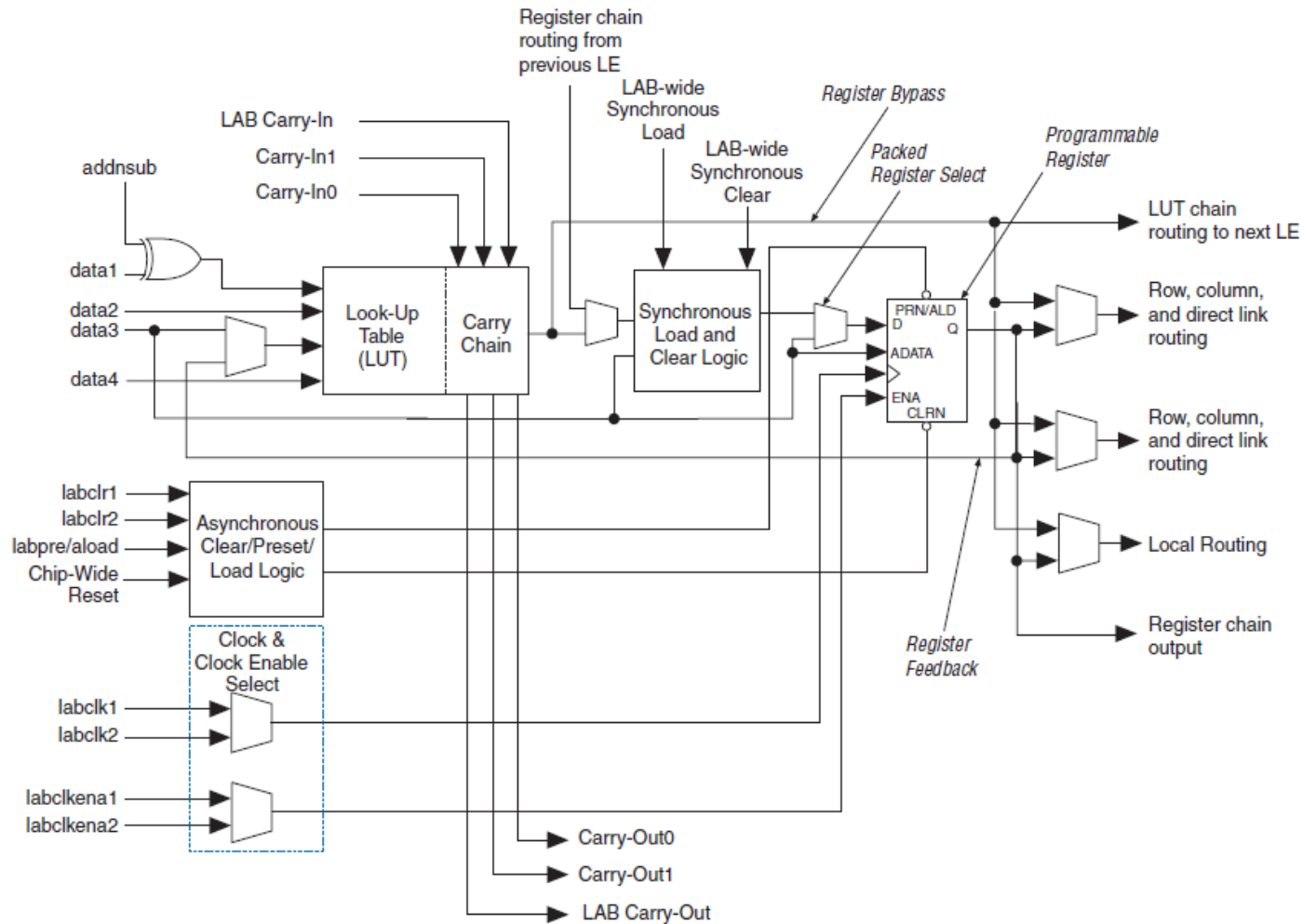
- example: ALTERA Cyclone



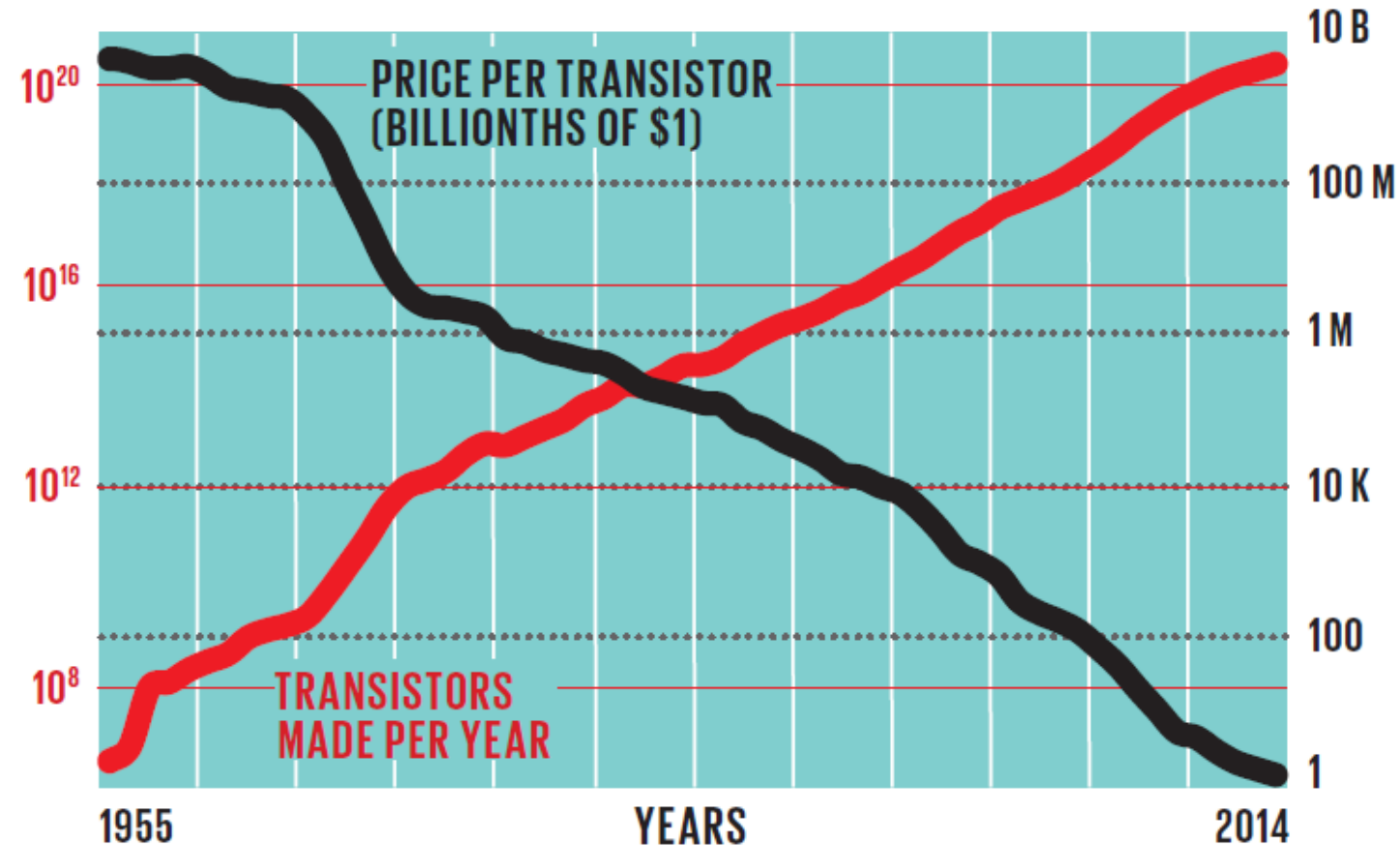
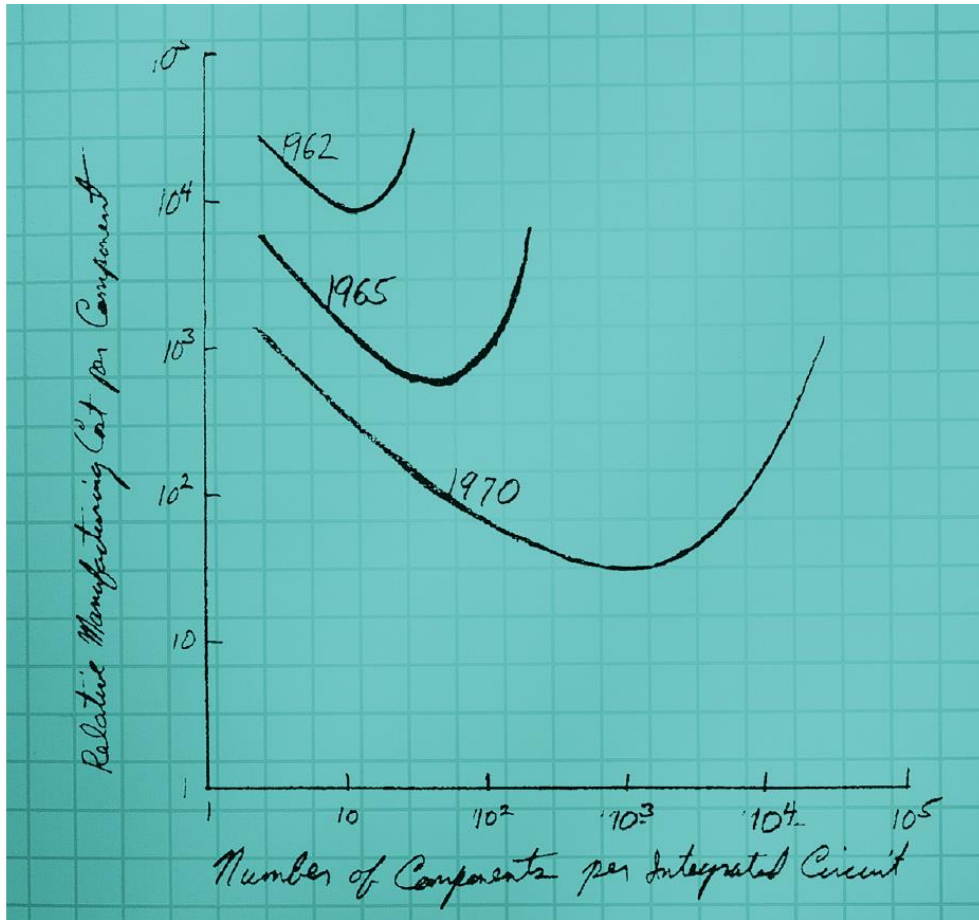
FPGA (2)

- example: ALTERA Cyclone

Logic element:



Scaling and Moore's law (1)

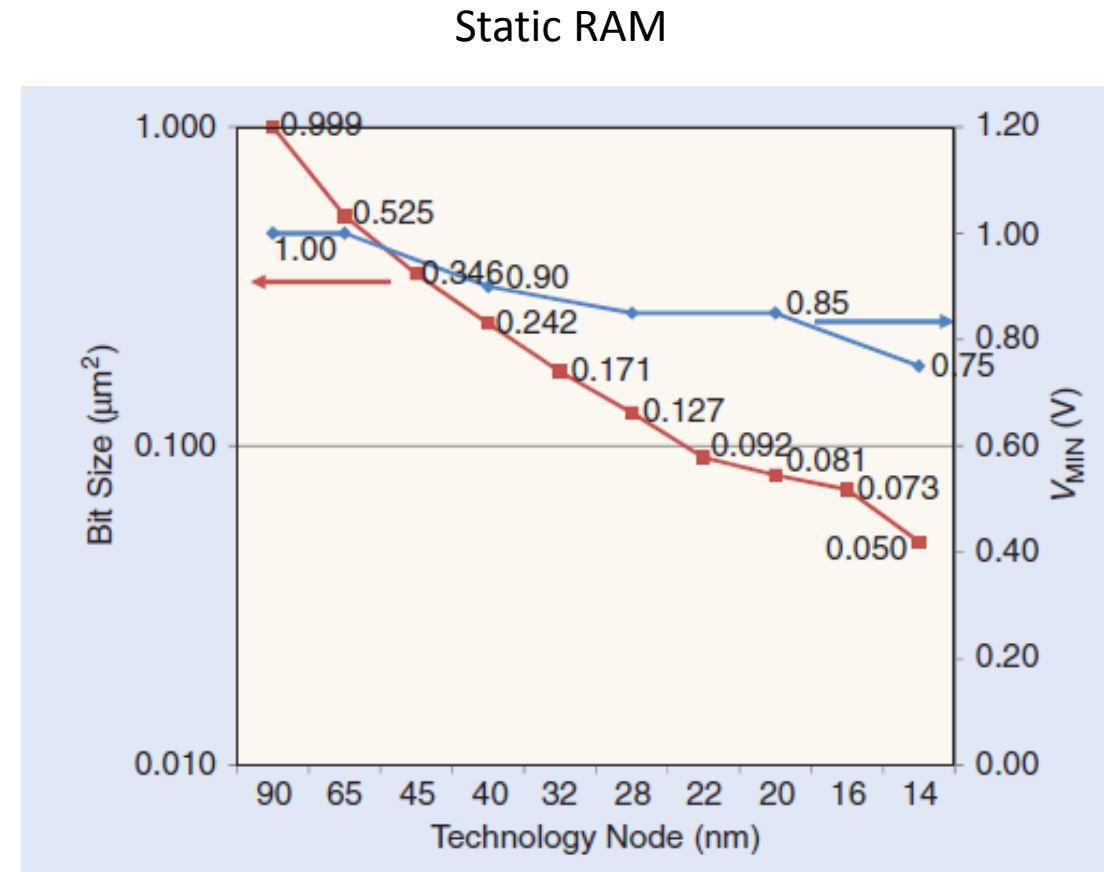
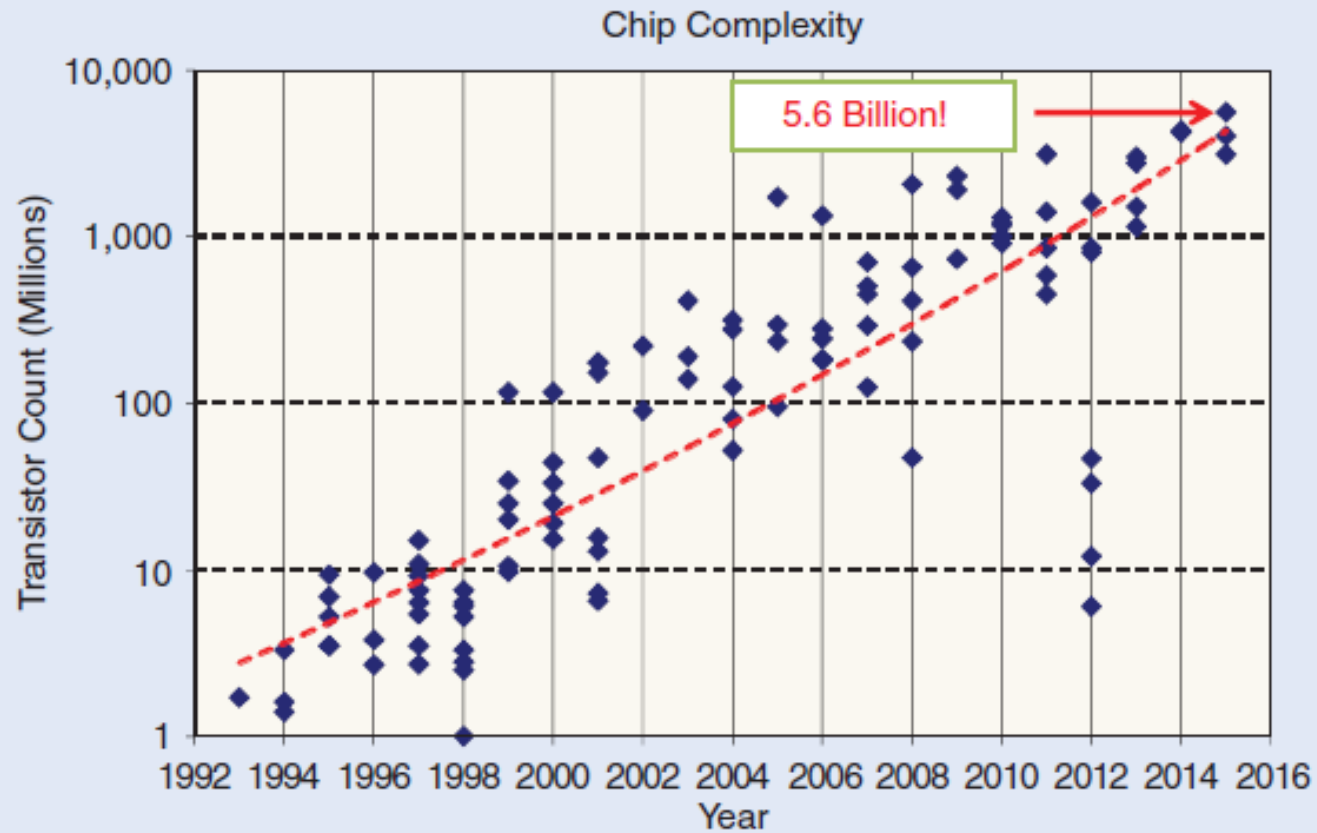


Scaling and Moore's law (2)

- 2014: 250×10^{18} transistors fabricated
- more than all transistors fabricated before 2010
- fabrication costs $\sim 1\text{B}\$/\text{acre}$
- cost of the tools: 100 times higher over 35 years
 - but 100 times faster....

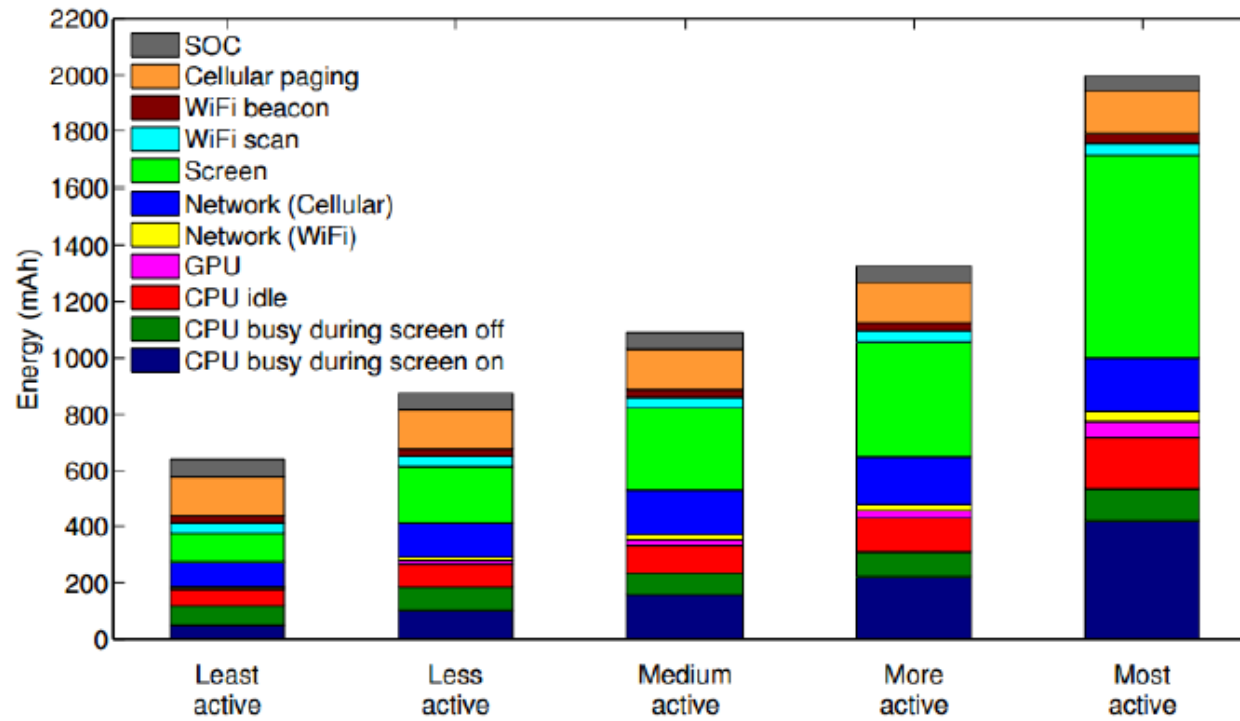
1 acre =
4 046.85642 m²

Scaling and Moore's law (3)



Power/energy consumption (1)

- Samsung Galaxy S3 – S4
 - Average energy drain breakdown of 1520 users (classified by activity rate)
 - X. Chen et al., “Smartphone Energy Drain in the Wild: Analysis and Implications”, ACM SIGMETRICS 2015

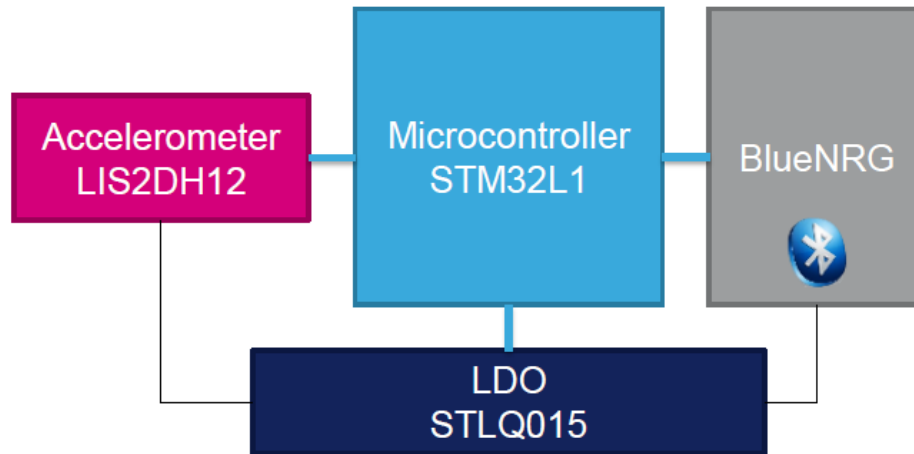


Power/energy consumption (2)

Typical energy breakdown [cont.] 17

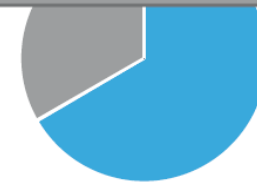
Wireless sensor node by STMicroelectronics

- **Microcontroller** (STM32L1): 1 μ A (stop mode w/RTC), 195 μ A/MHz Run mode (6mA @32MHz)
- **Accelerometer** (LIS2DH12): 2 μ A @1Hz, 11 μ A @50Hz, 0.5 μ A power-down
- **Bluetooth LE network processor** (BlueNRG): down to 1.7 μ A with active BLE stack, 8.2 mA maximum TX current (@0 dBm, 3.0 V)
- **LDO regulator** (STLQ015): quiescent current ~1.4 μ A, 1 nA in OFF mode

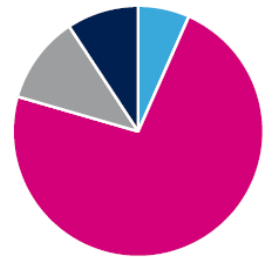


Normal mode (~10mA)	Low power mode (~8 μ A)
CPU running @32MHz	CPU in stop mode w/RTC
Accelerometer sampling @100Hz	Accelerometer sampling @50Hz,
Bluetooth LE transmitting @50Hz	Low-power mode, active BLE stack
LDO on	LDO on

The system is in passive mode and is periodically woken up by clock or by external event (e.g., free fall detection)

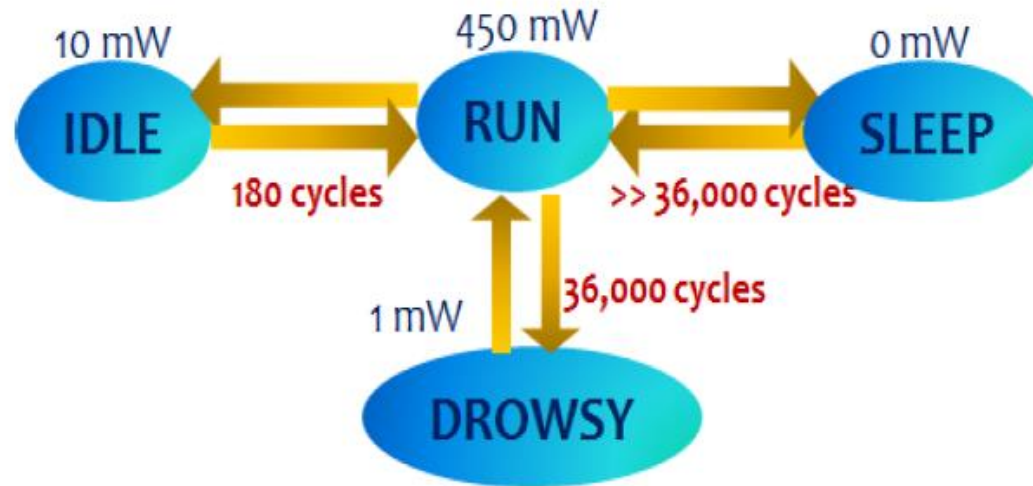


Low-power mode



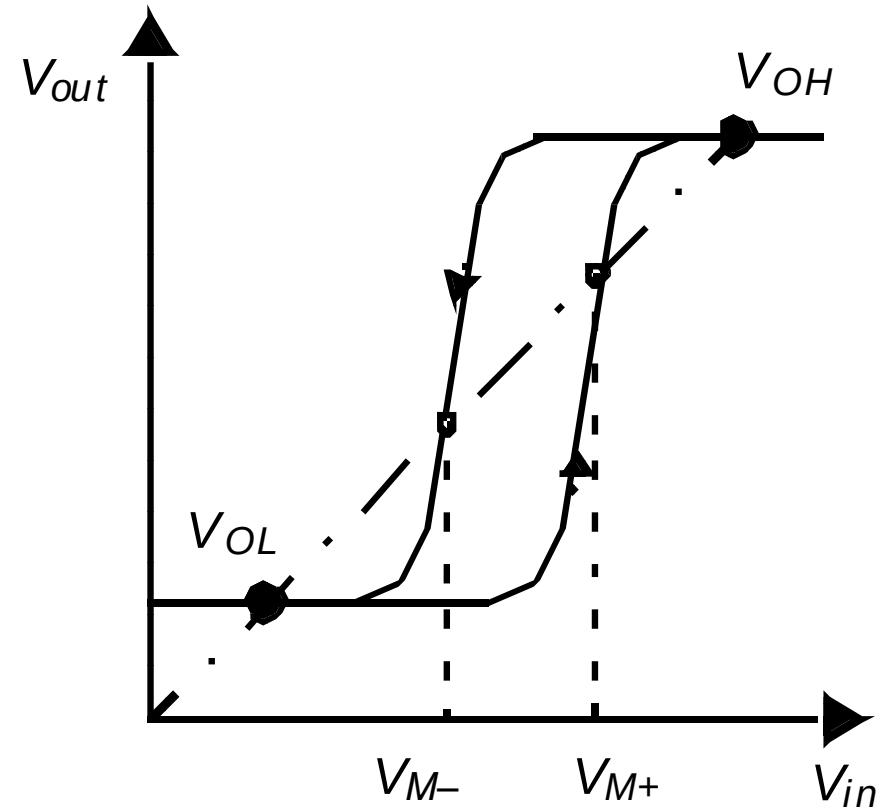
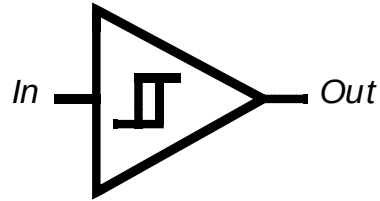
Power/energy consumption (3)

- **Power state machine**
 - States = operative modes
 - Transitions have power/performance costs
- Example (Intel Xscale core)



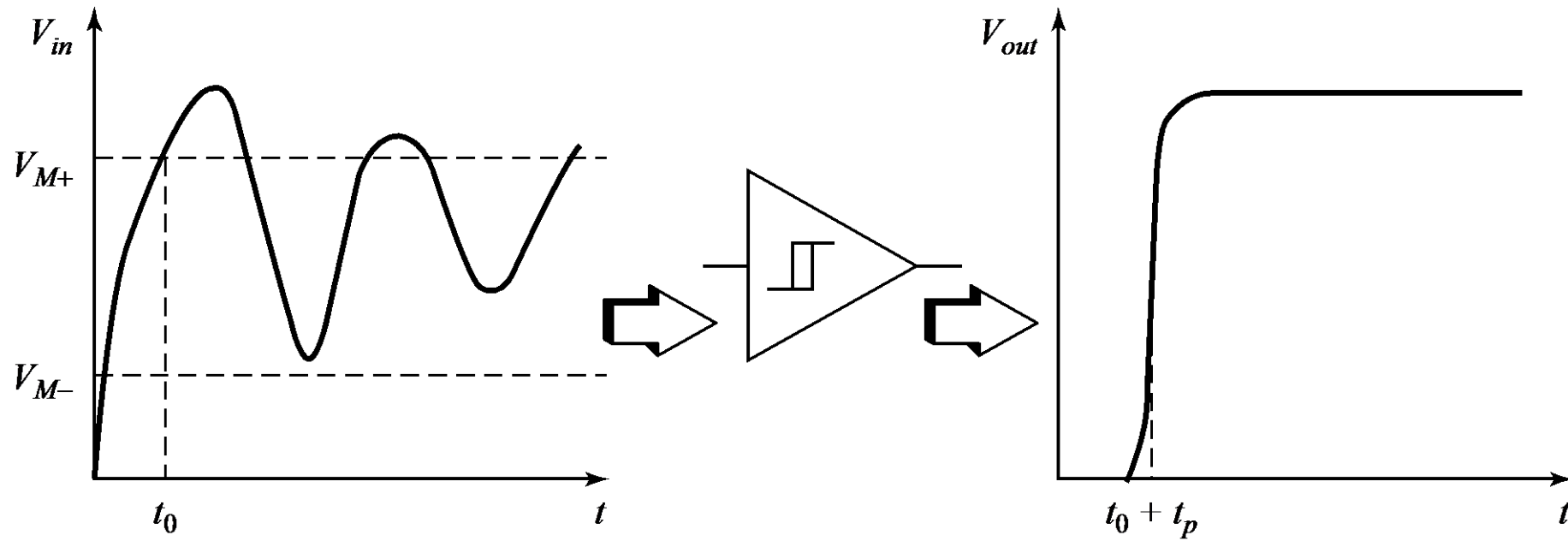
E. Macii, M. Grosso, M. Poncino
Power Efficiency in the Design of IoT Devices

Schmitt Trigger (1)

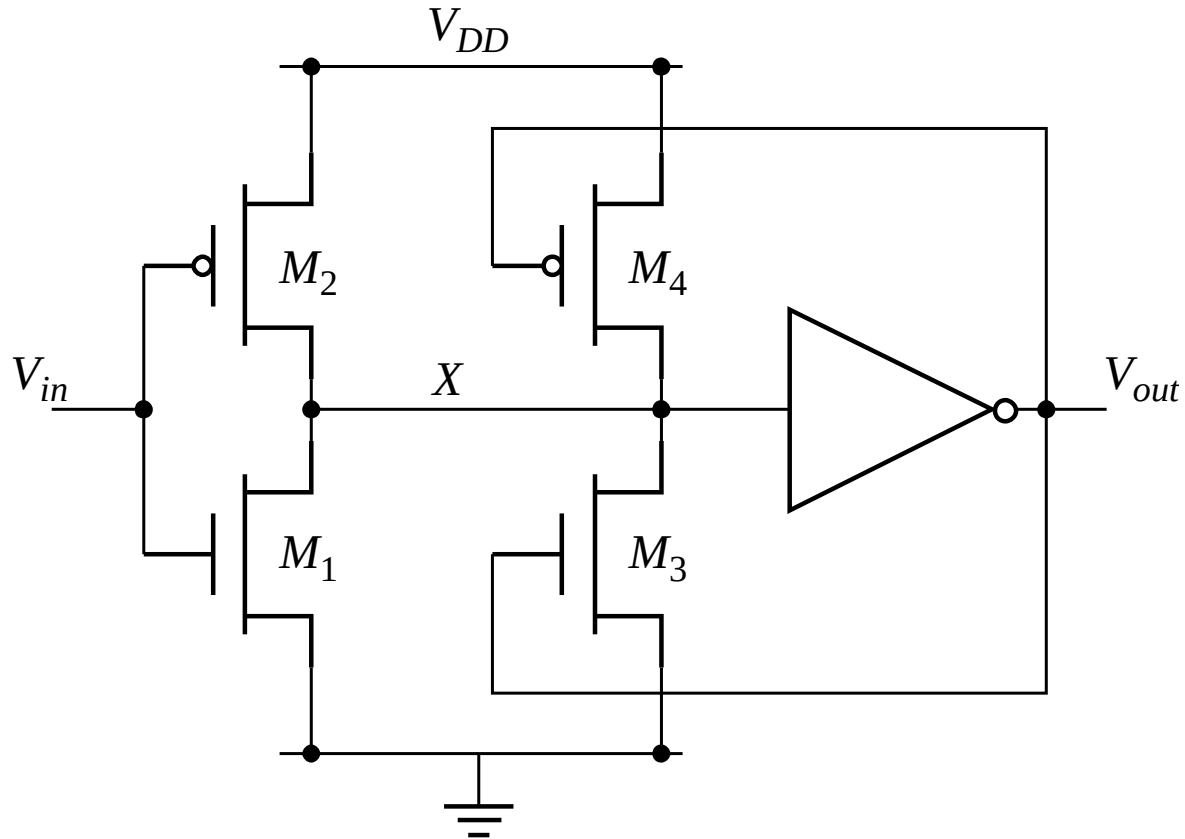


- bistable circuit use to restore noisy signals

Schmitt Trigger (2)



Schmitt Trigger (3)

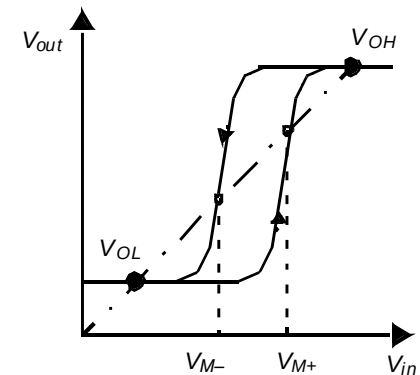


M2+M4 if "out"='0'
M2 if "out"='1'

$$V_{LT} = \frac{V_{DD} \sqrt{\frac{\beta_{up}}{\beta_{down}}} - V_T \left(\sqrt{\frac{\beta_{up}}{\beta_{down}}} - 1 \right)}{1 + \sqrt{\frac{\beta_{up}}{\beta_{down}}}}$$

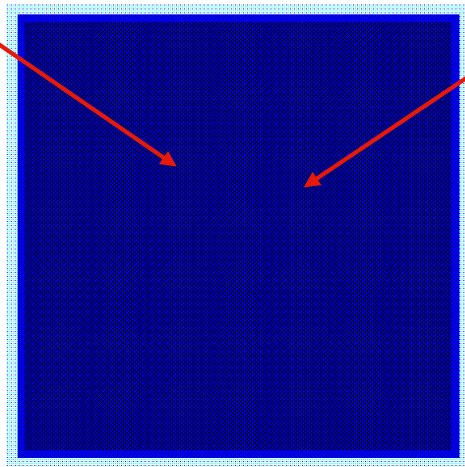
M1+M3 if "out"='1'
M1 if "out"='0'

$$V_{LT}(\text{"out"}='1') < V_{LT}(\text{"out"}='0')$$



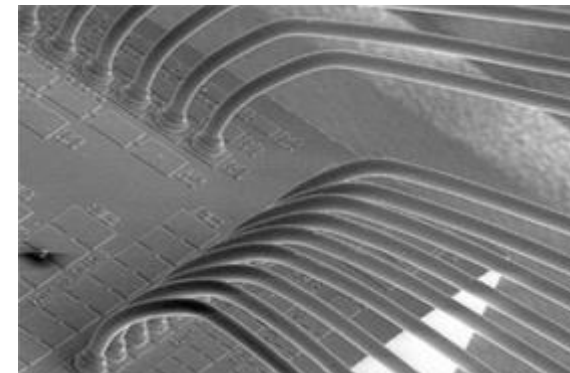
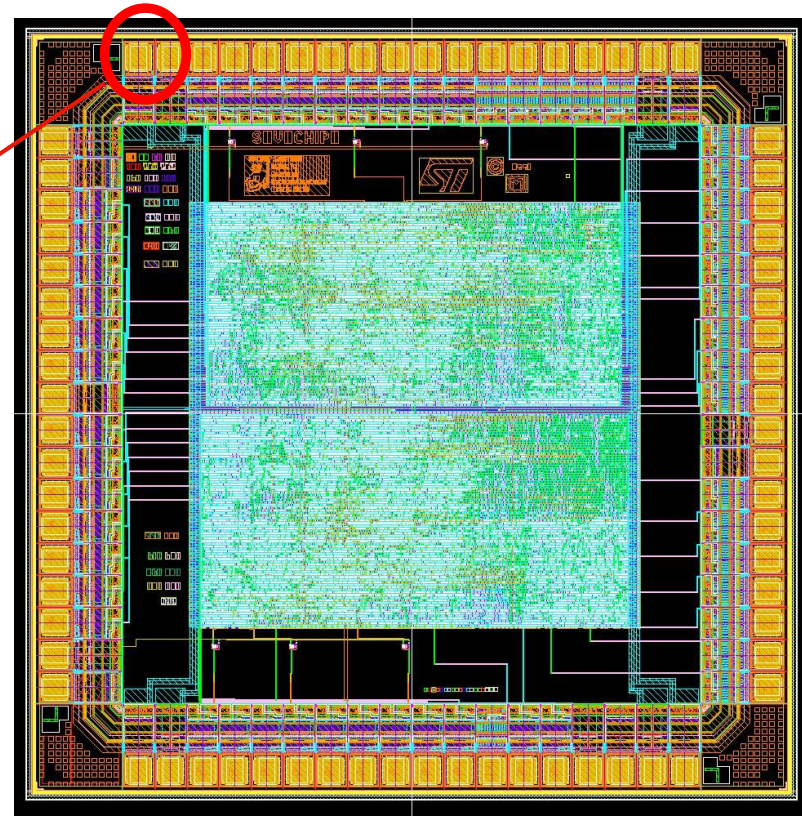
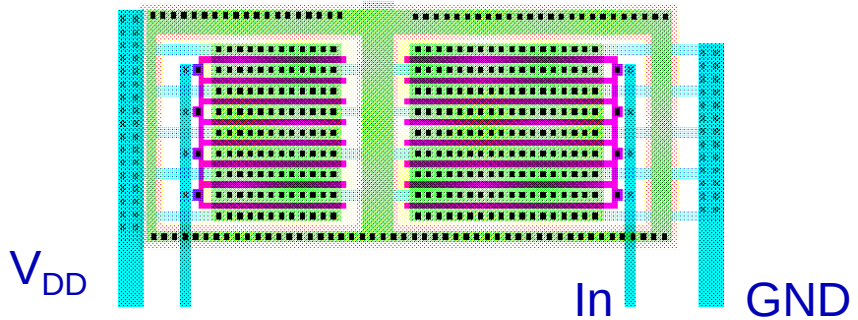
Pads (1)

Bonding Pad



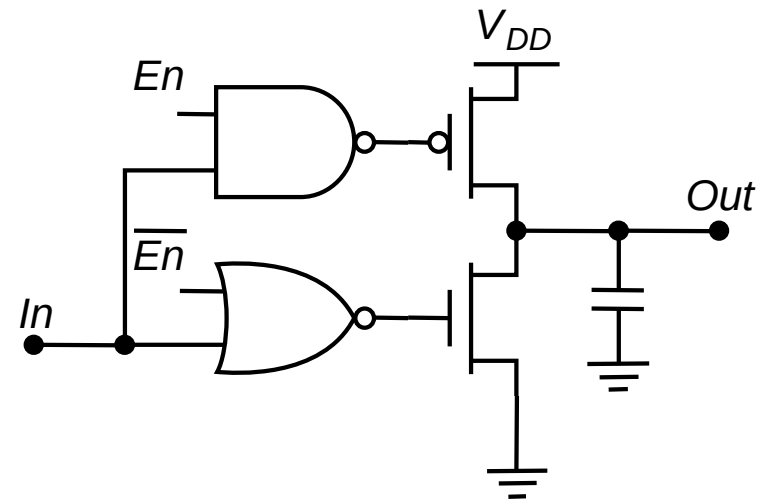
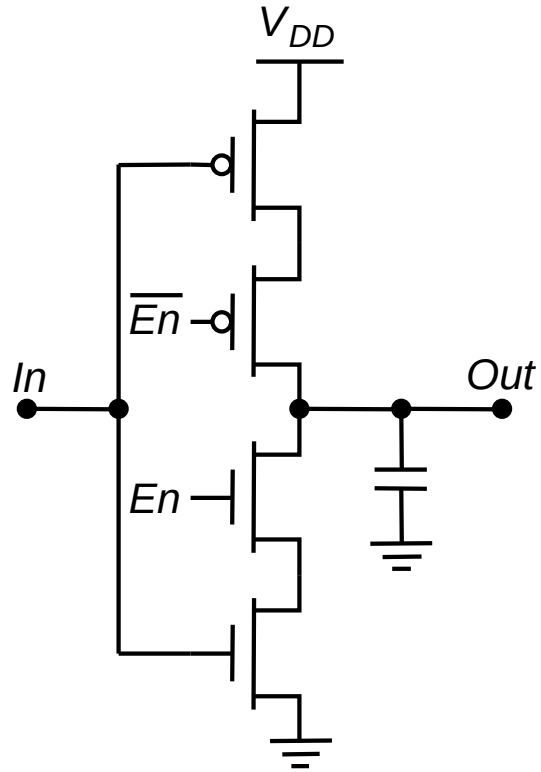
100 μm

Out



PADS +
bonding
wires

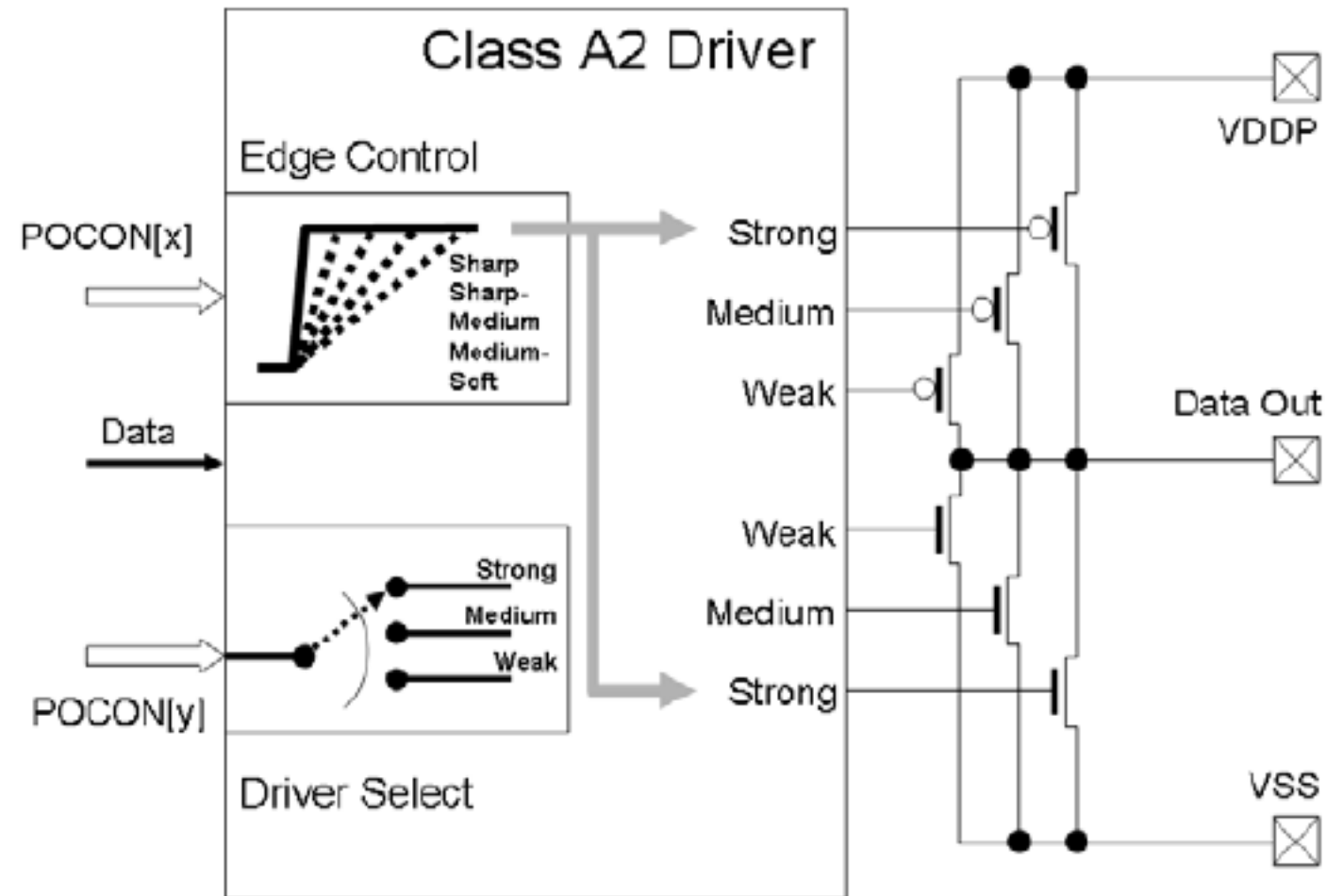
Pads (2): tri-state buffer



Increased output drive

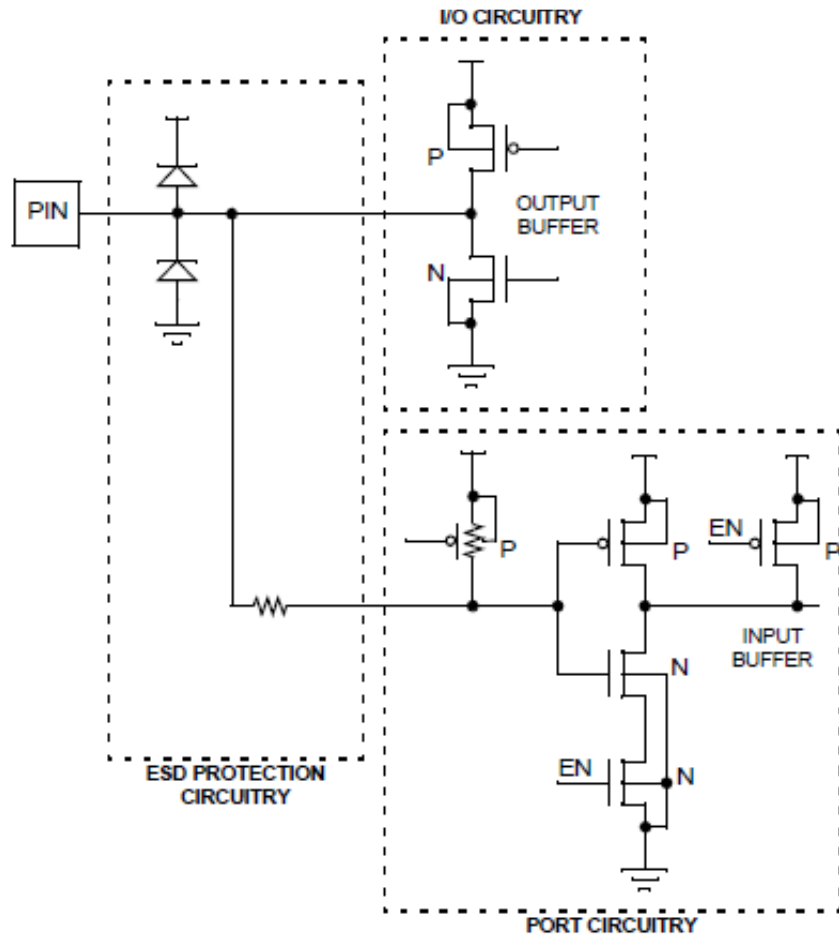
Pads (3)

- scalable pads (Infineon AP32146)

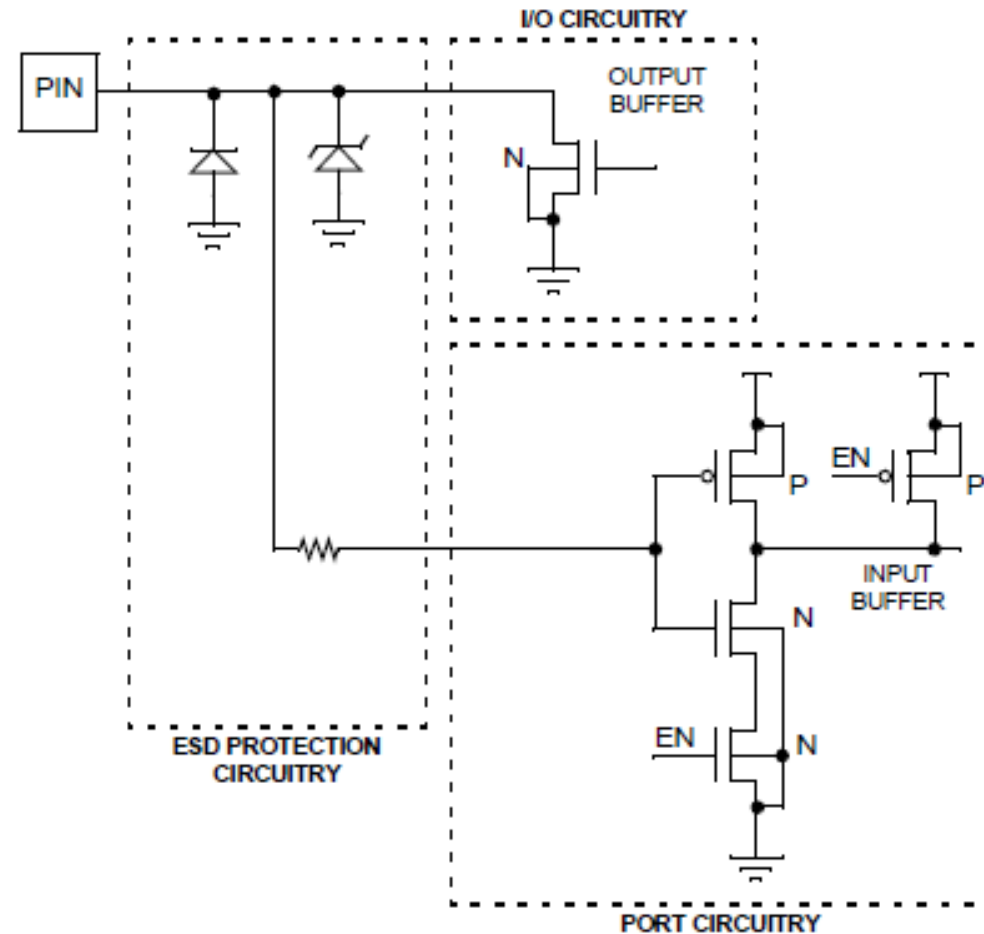


Example of Pads in ST microcontrollers (1)

- digital push-pull

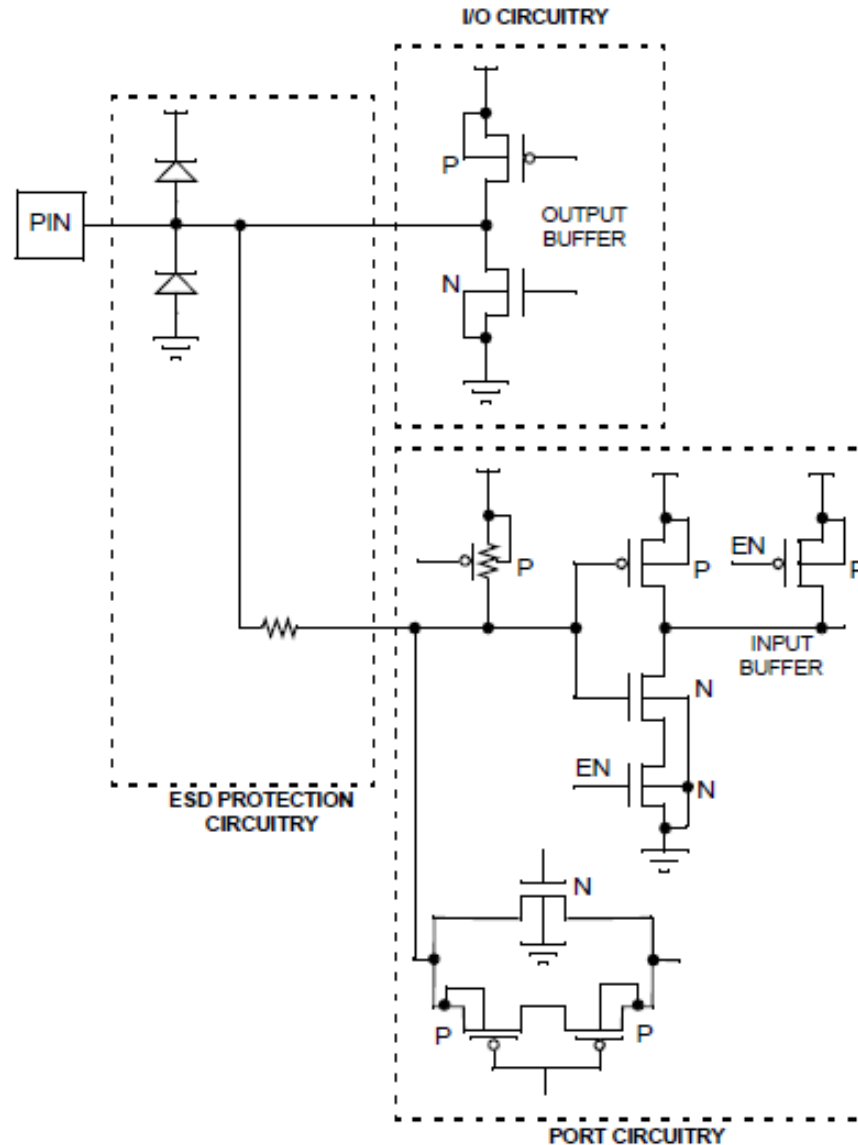


- open drain

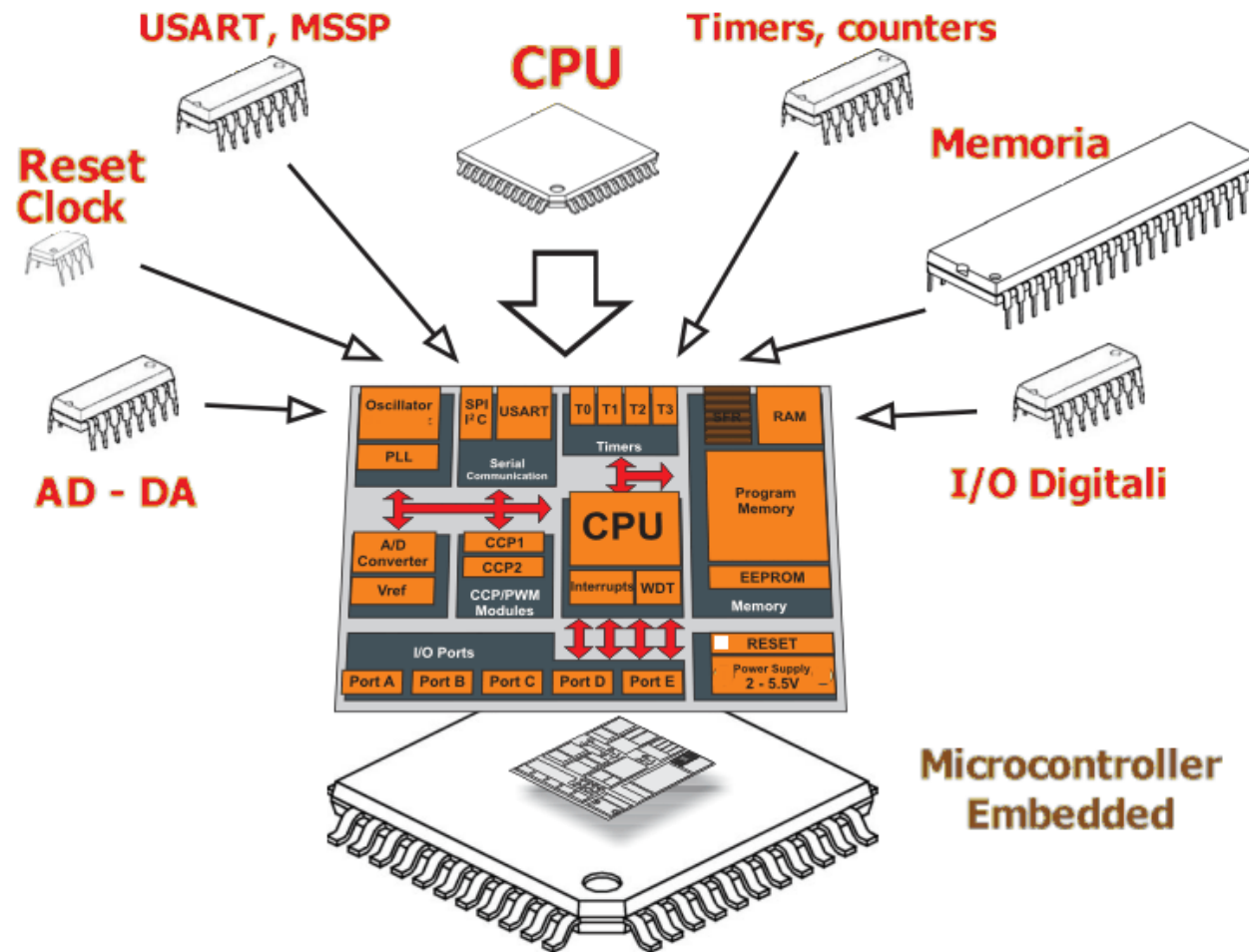


Example of Pads in ST microcontrollers (2)

- with analog multiplexer



Microcontrollers (1)



thanks to N.Ponte

Microcontrollers (2)

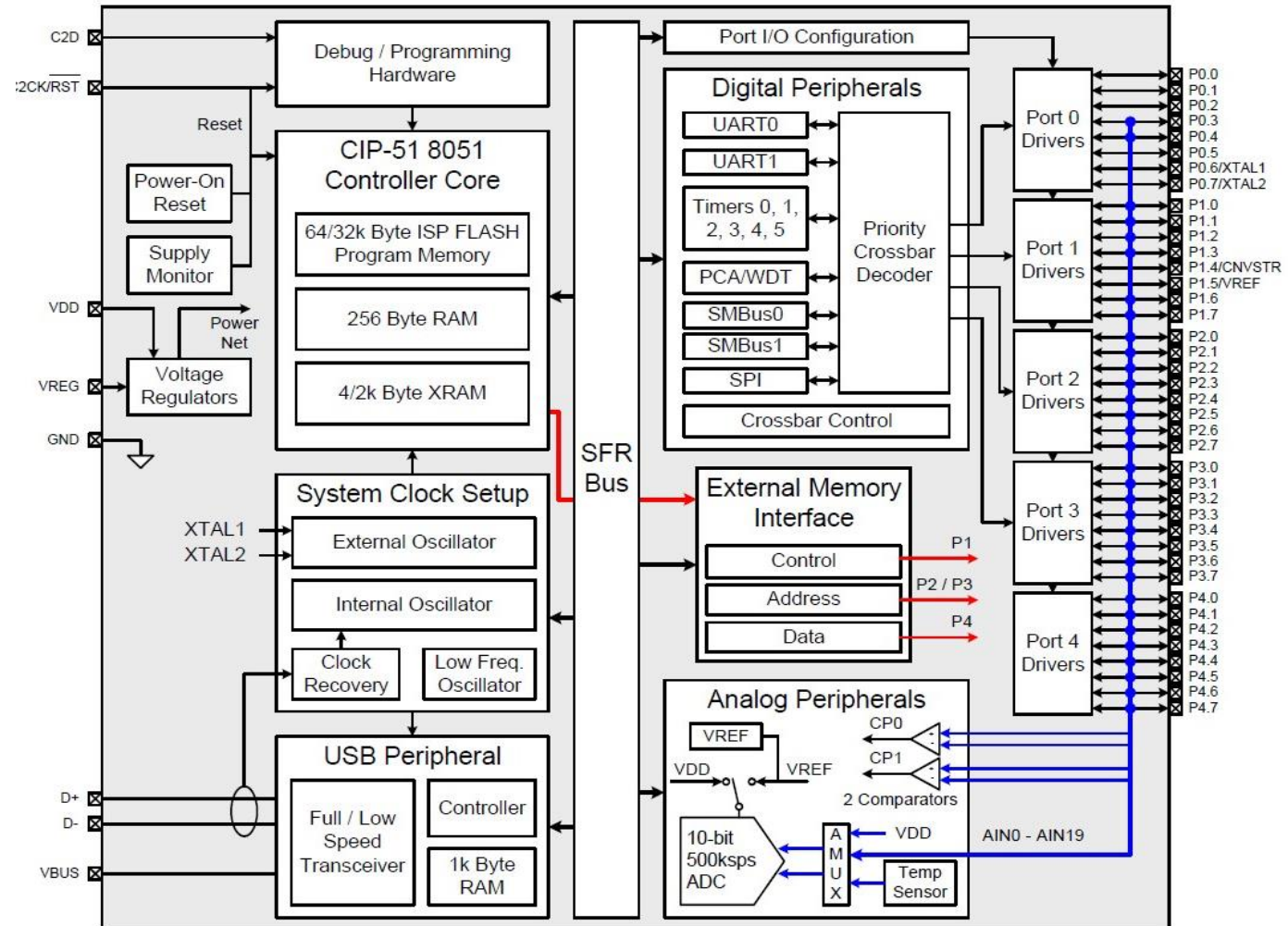
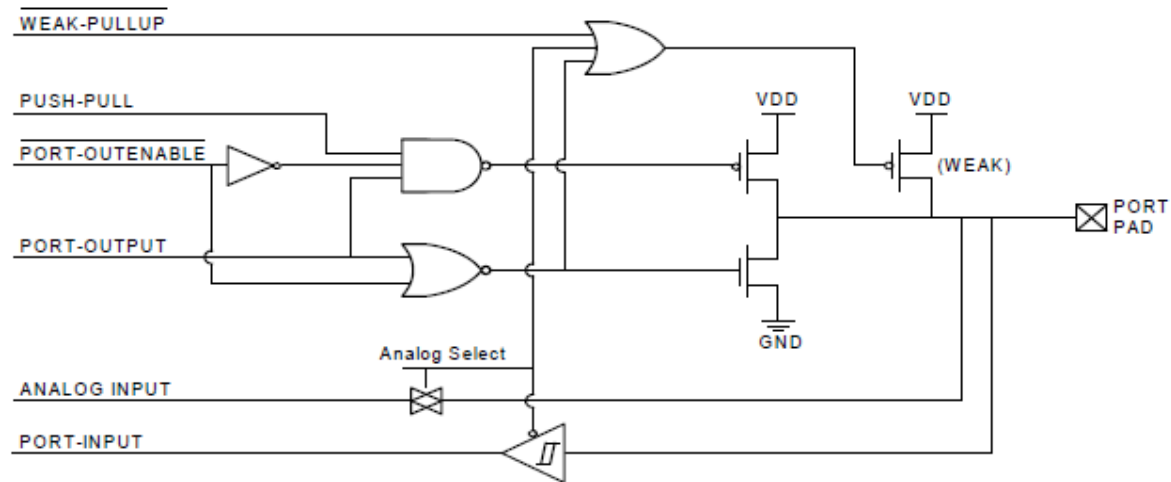


Figure 1.1. C8051F380/2/4/6 Block Diagram

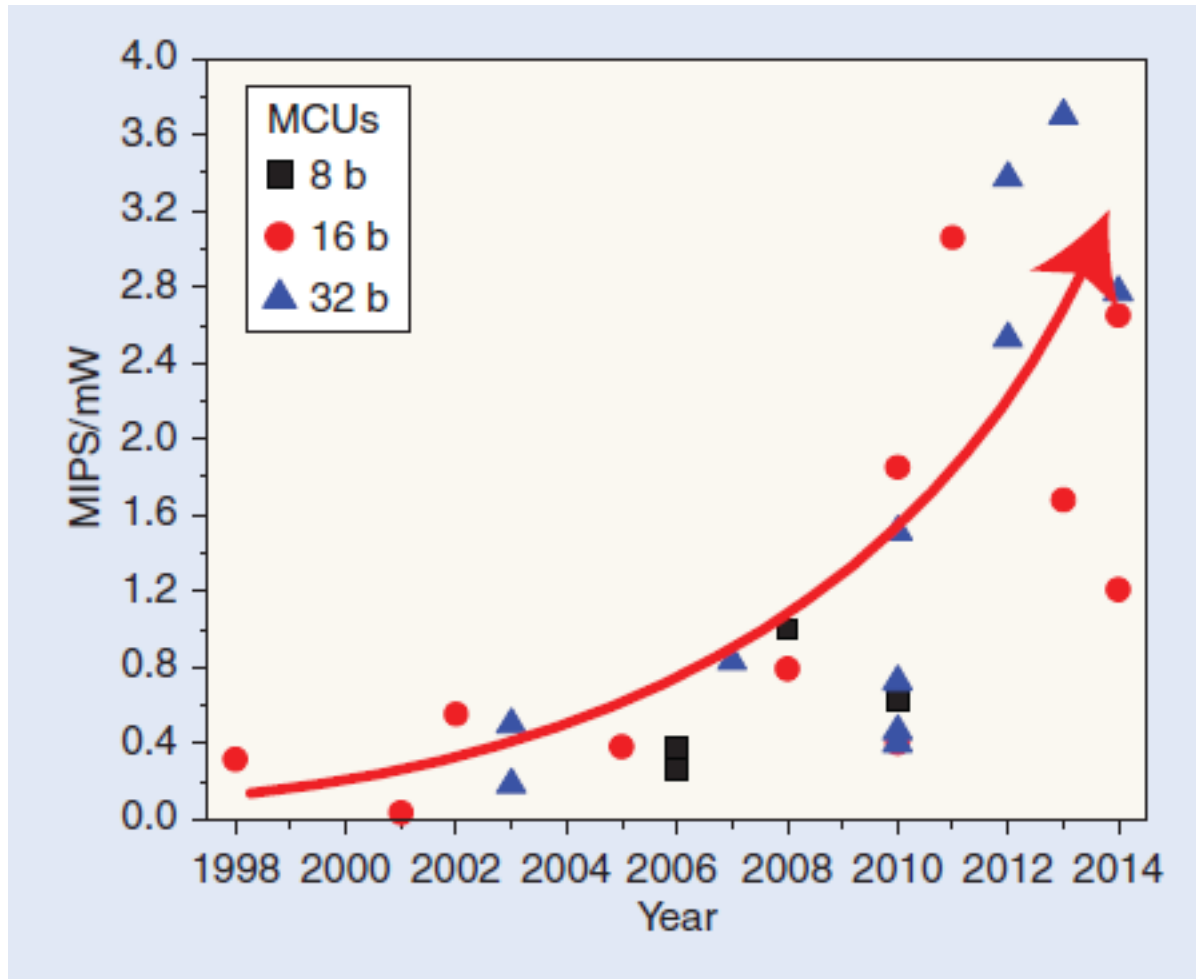
thanks to N.Ponte

Microcontrollers (3)

I/O of C8051F380 microcontroller

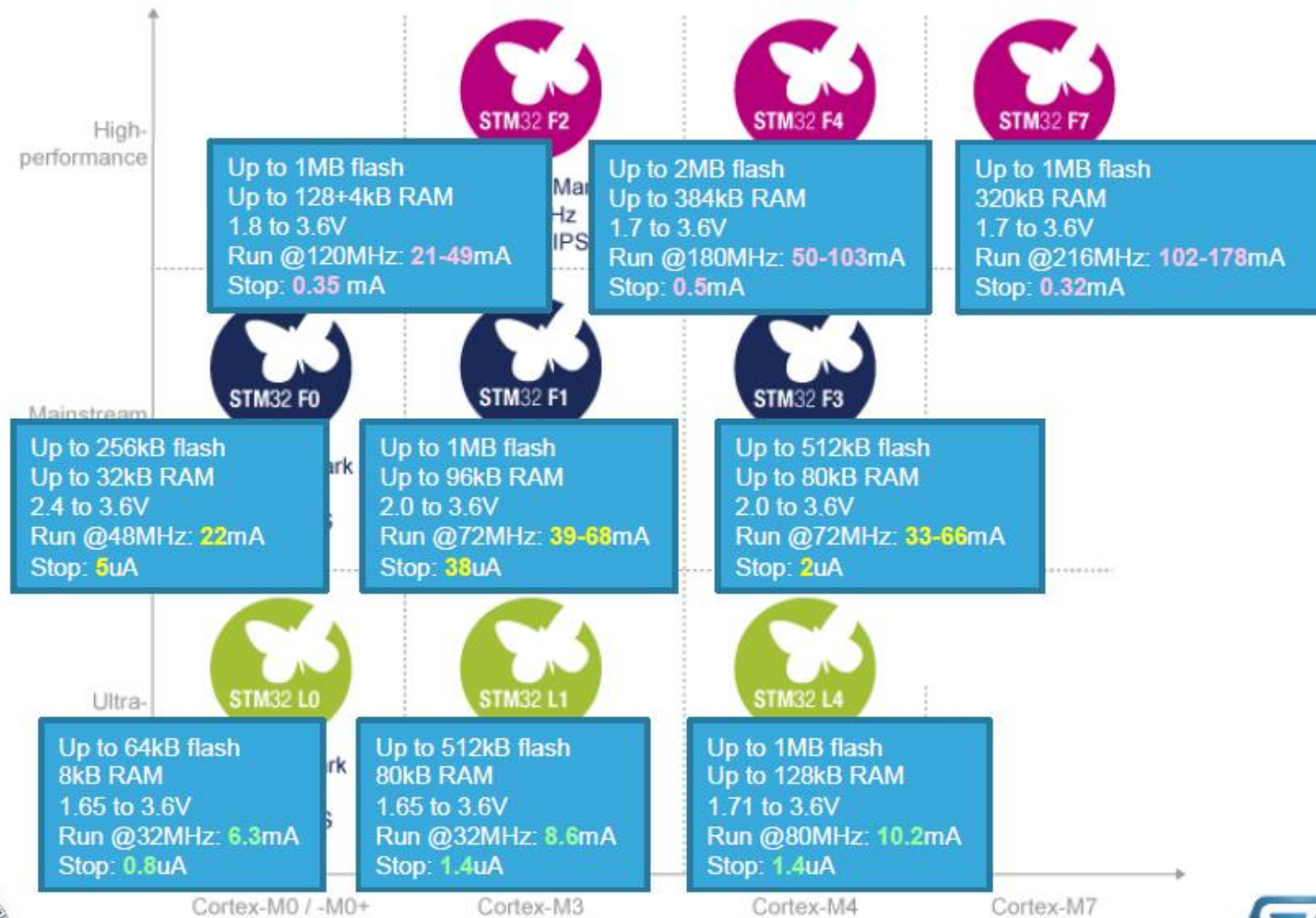


Microcontrollers (4)



huge efforts to improve
energy efficiency

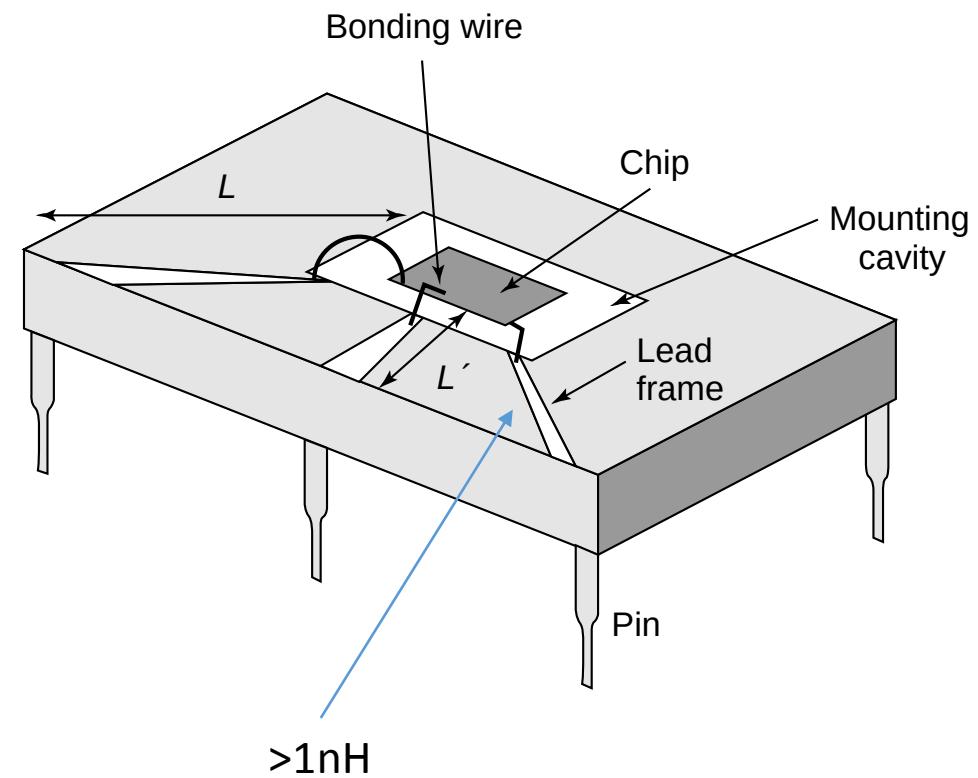
Microcontrollers (5)



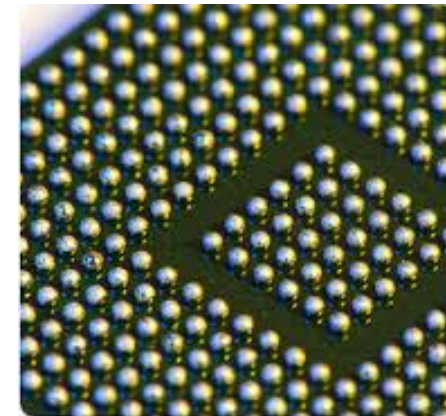
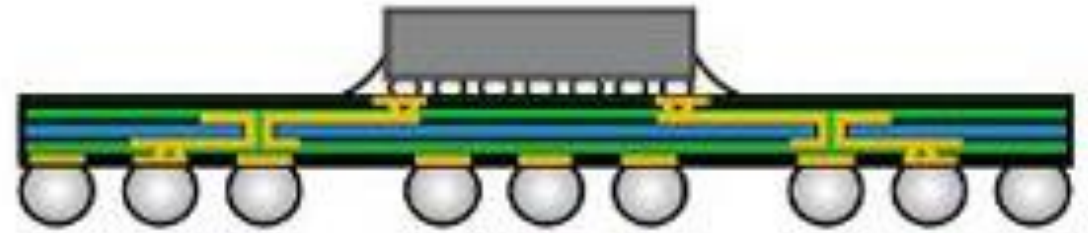
E. Macii, M. Grosso, M. Poncino
Power Efficiency in the Design of IoT Devices

Package (1)

- lead frame

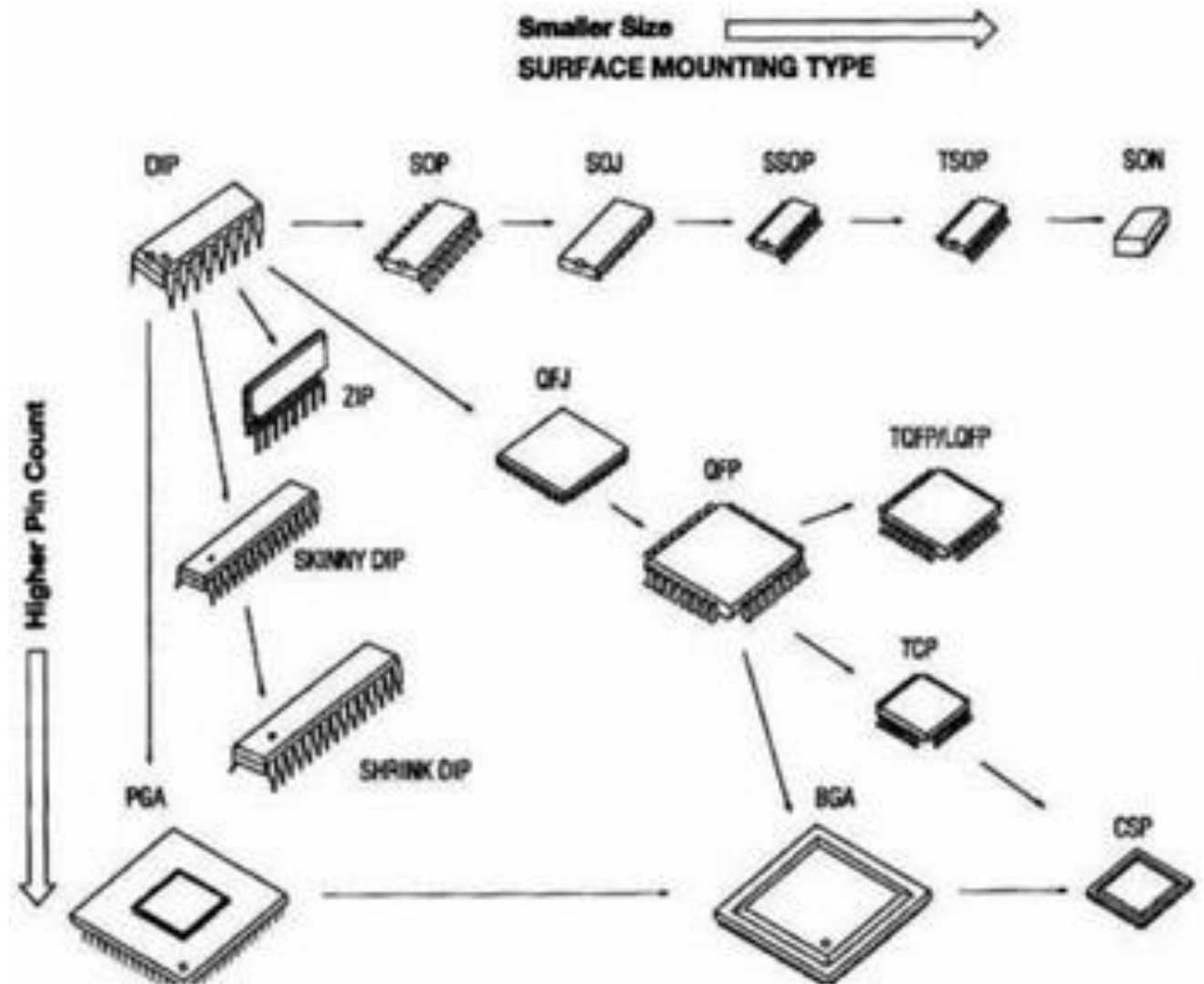
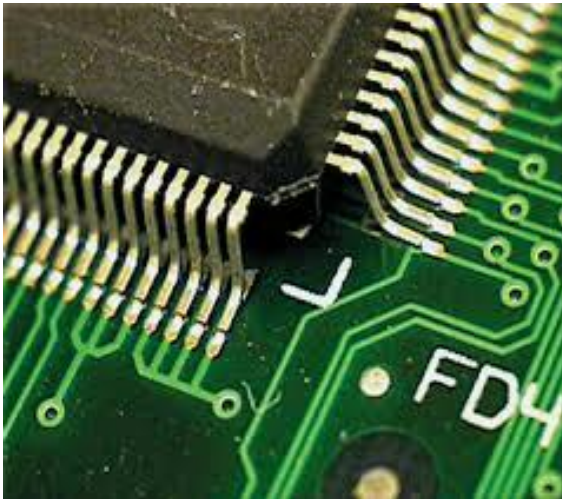


- flip-chip with ball-grid-array



Package (2)

SMD: surface mount device



PCB design

- AN203 Silicon Labs

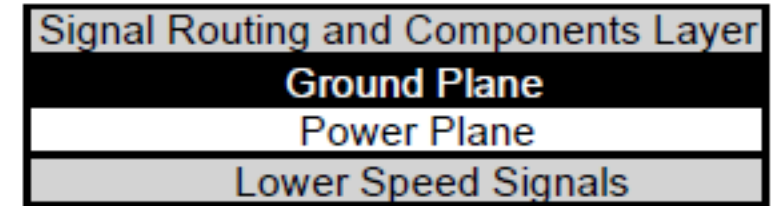
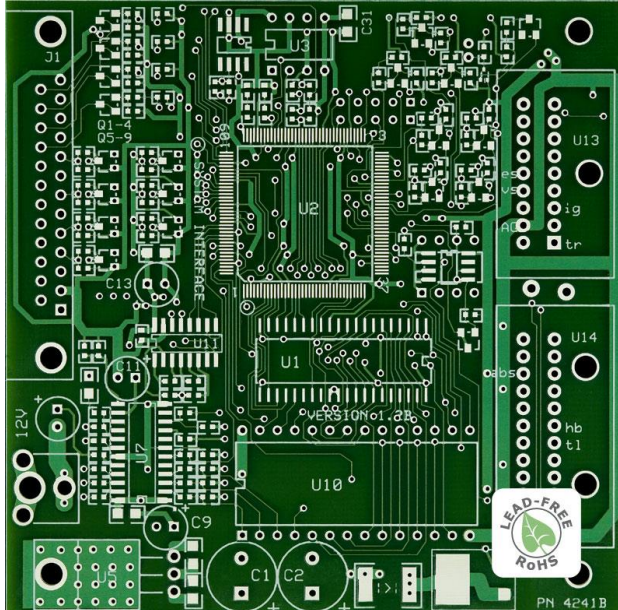


Figure 13. Four Layer Board

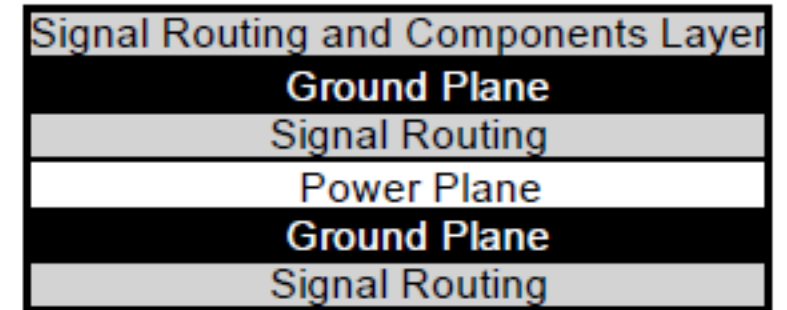


Figure 14. Six Layer Board

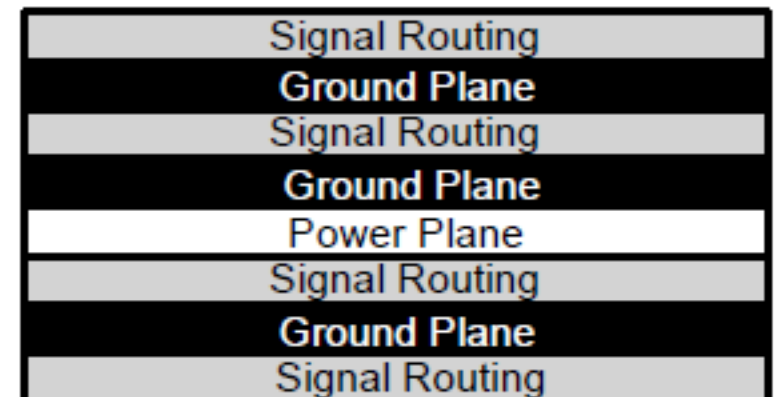
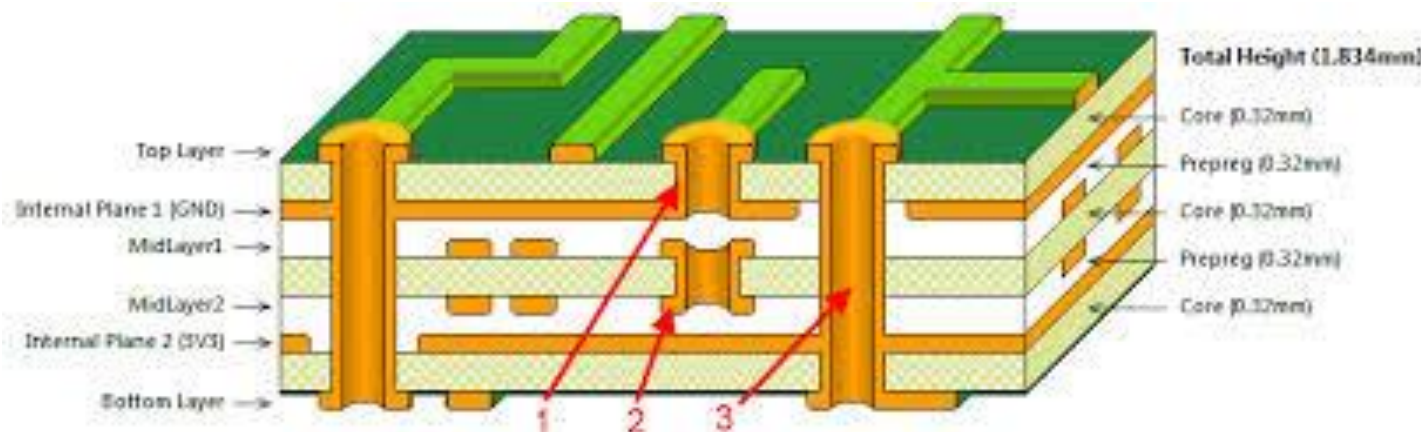


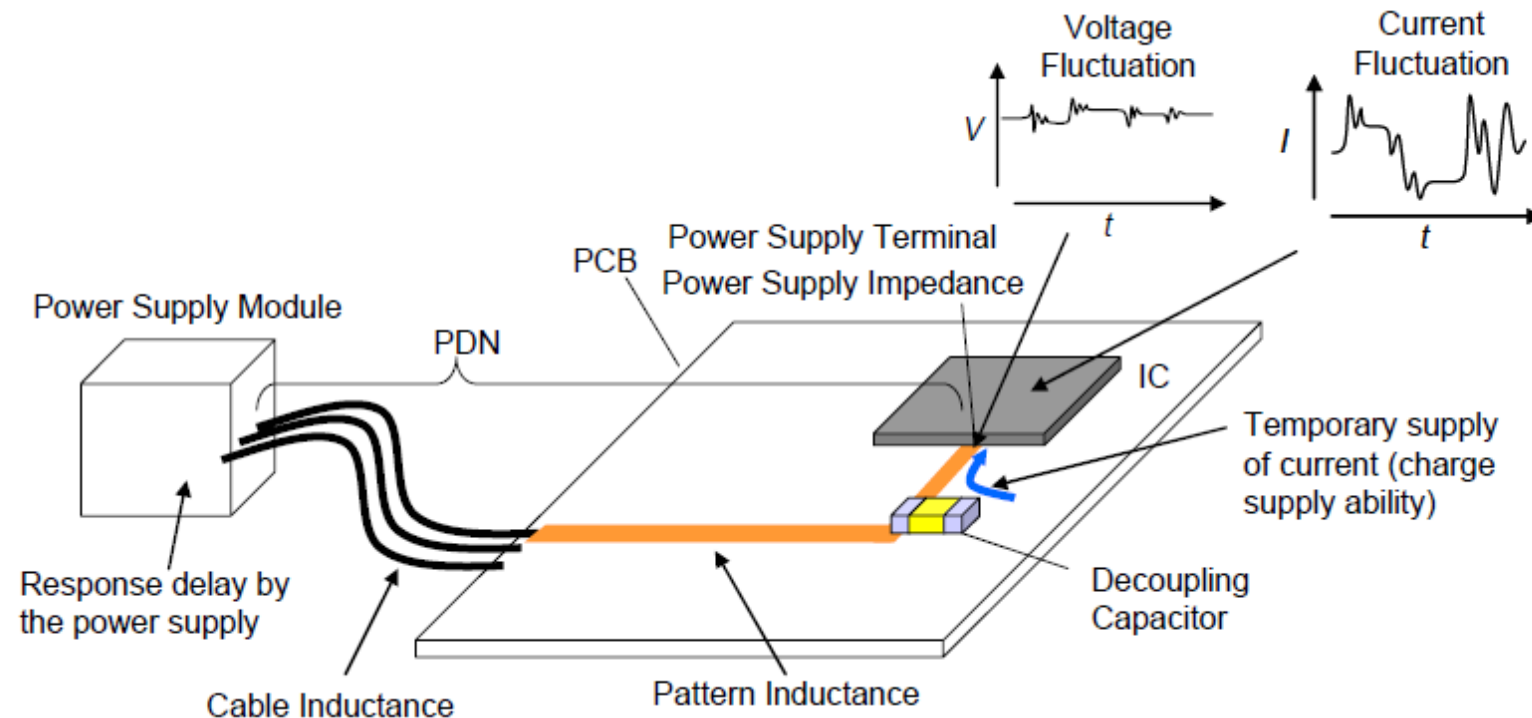
Figure 15. Eight Layer Board

PCB design: decoupling (1)

From:

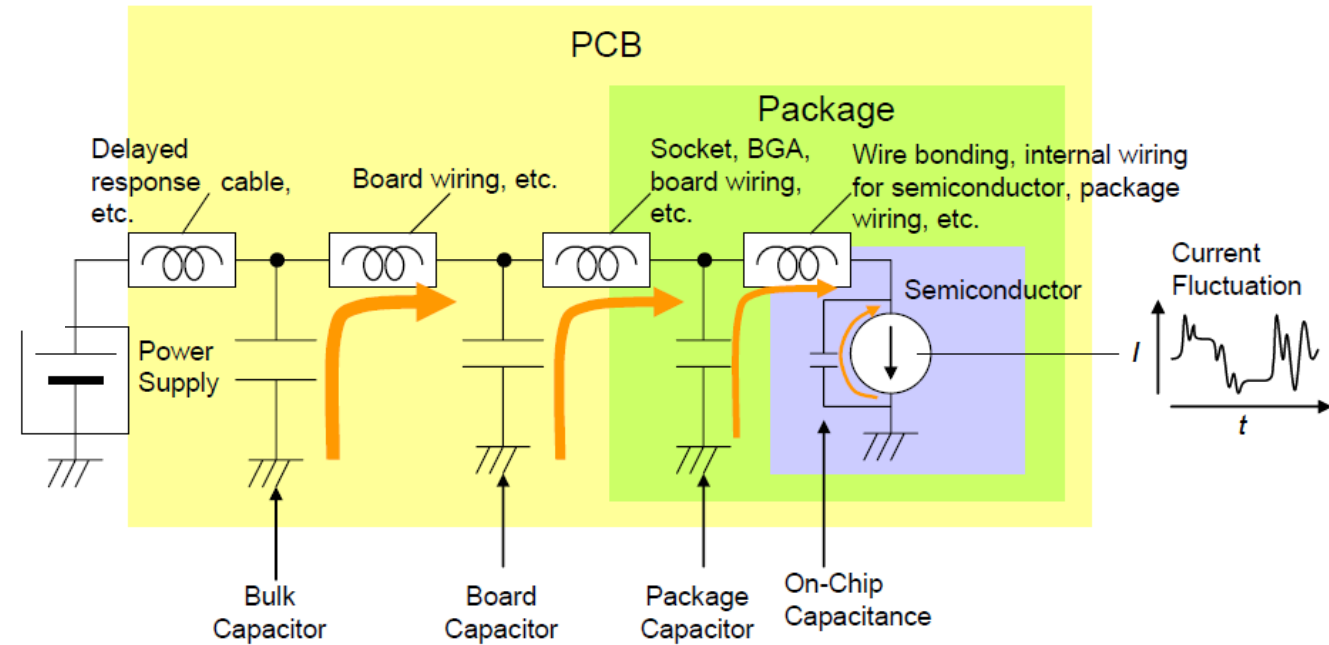
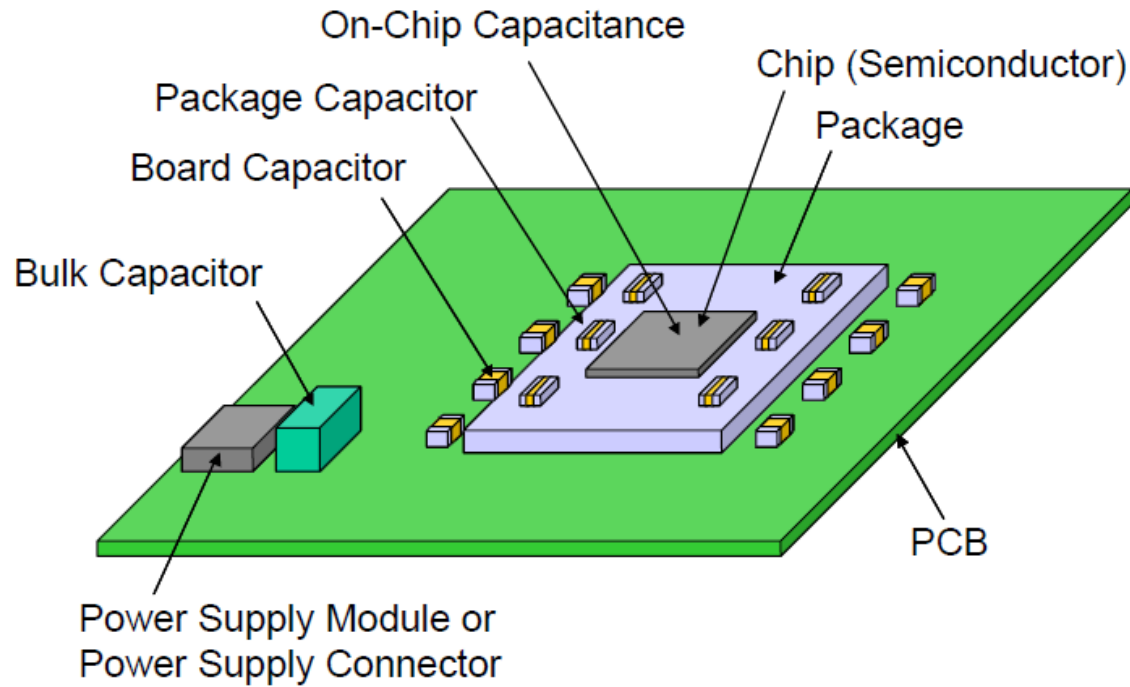
Application Manual
for Power Supply Noise Suppression
and Decoupling for Digital ICs

C39E.pdf
Jul.20,2010

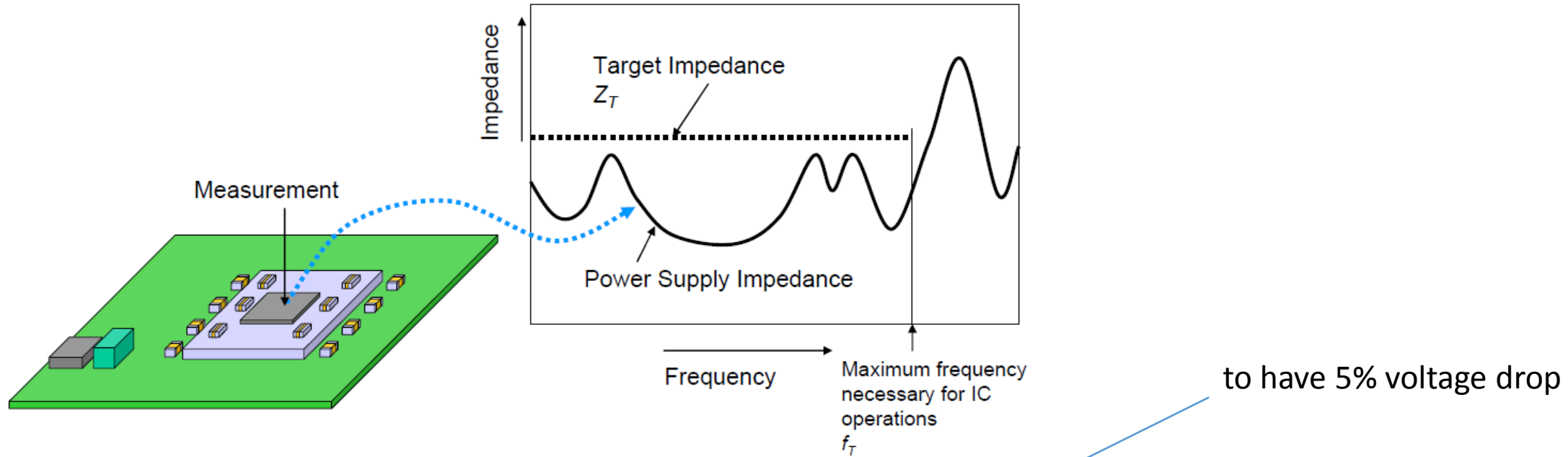


PDN: power distribution network

PCB design: decoupling (2)



PCB design: decoupling (3)

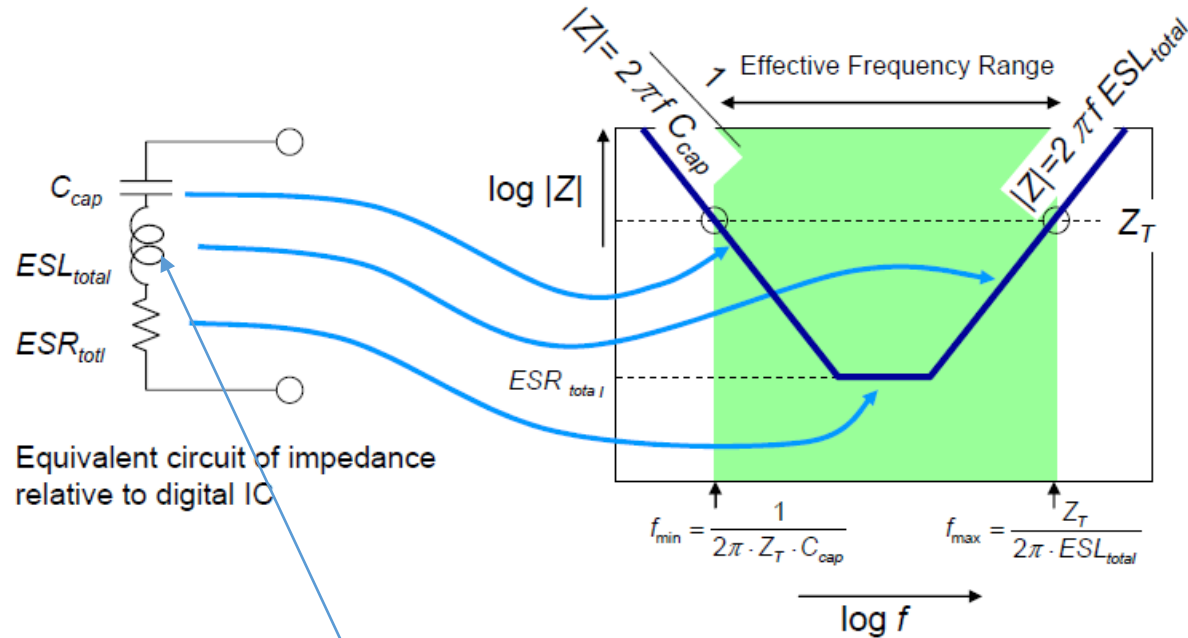


year	voltage (volts)	power dissipated (watts)	current (amps)	Z_{target} (mOhms)	frequency (MHz)
1990	5.0	5	1	250	16
1993	3.3	10	3	54	66
1996	2.5	30	12	10	200
1999	1.8	90	50	1.8	600
2002	1.2	180	150	0.4	1200

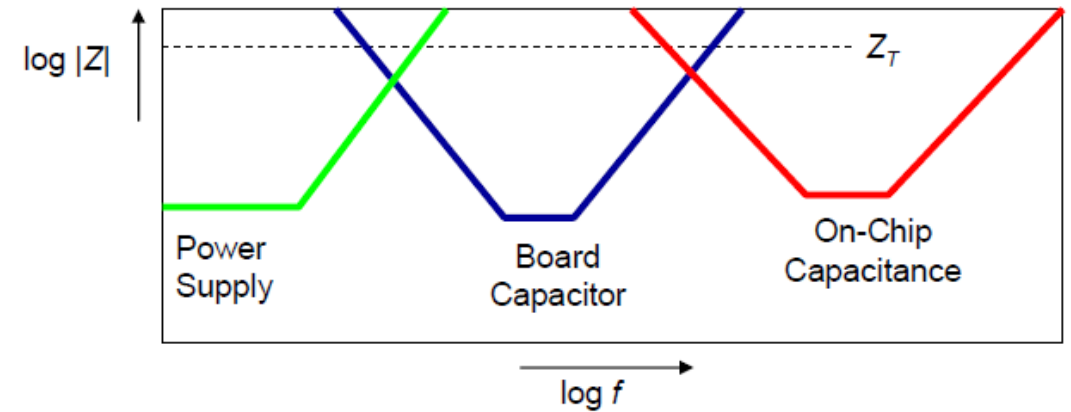
Smith et al., IEEE
TRANSACTIONS ON
ADVANCED PACKAGING,
VOL. 22, NO. 3, AUGUST
1999

PCB design: decoupling (4)

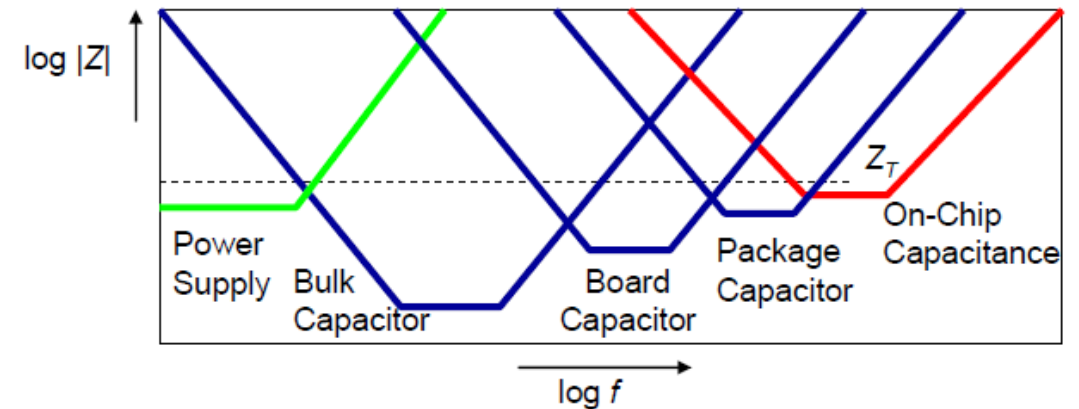
impedance vs. freq. of a single capacitor



not only the capacitor itself,
but also the trace inductance



(a) When target impedance is high



(b) When target impedance is low

PCB design: decoupling (5)

STEP 1

Establishing target impedance

$$Z_T = \frac{\Delta V}{\Delta I}$$

ΔV : Maximum allowable ripple voltage
 ΔI : Current Fluctuation (rated current when unknown)

STEP 2

Establishing maximum frequency f_T for IC power supply terminal
 100MHz when unknown

STEP 3

Establishing bulk capacitor capacitance

$$C_{bulk} \geq \frac{L_{Power}}{Z_T^2}$$

STEP 4

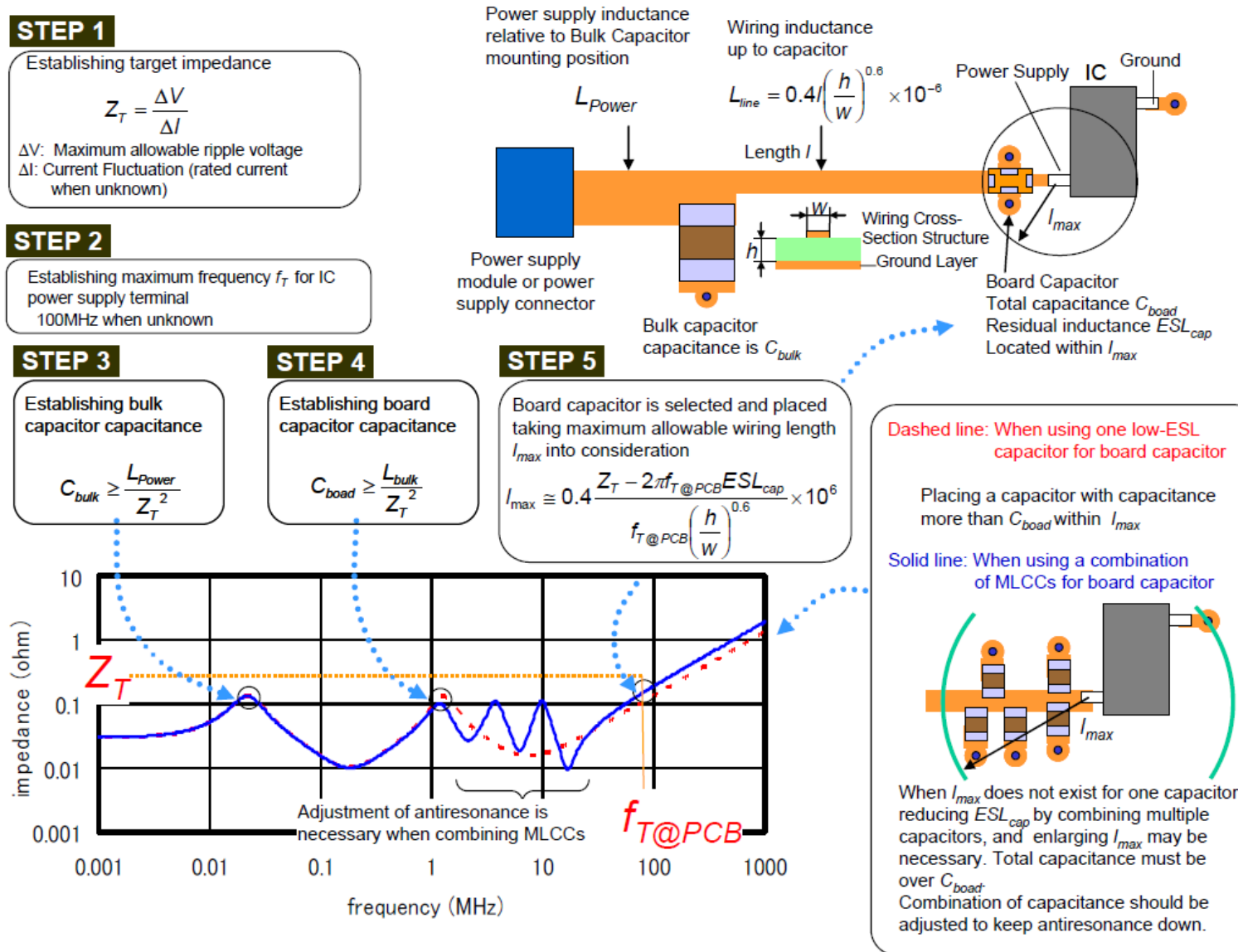
Establishing board capacitor capacitance

$$C_{board} \geq \frac{L_{bulk}}{Z_T^2}$$

STEP 5

Board capacitor is selected and placed taking maximum allowable wiring length l_{max} into consideration

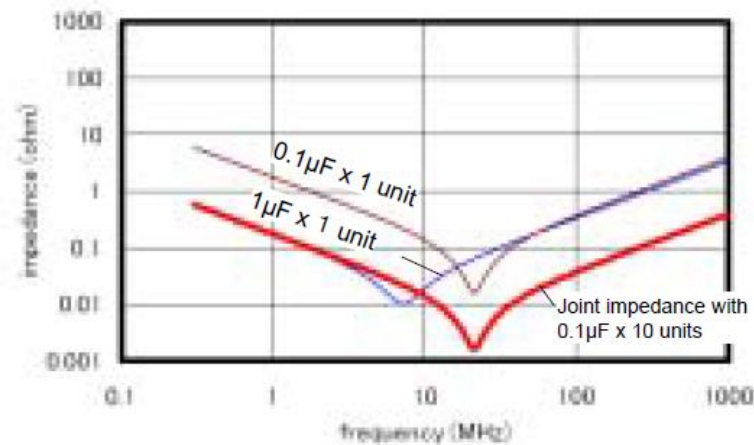
$$l_{max} \approx 0.4 \frac{Z_T - 2\pi f_{T@PCB} ESL_{cap}}{f_{T@PCB} \left(\frac{h}{W}\right)^{0.6}} \times 10^6$$



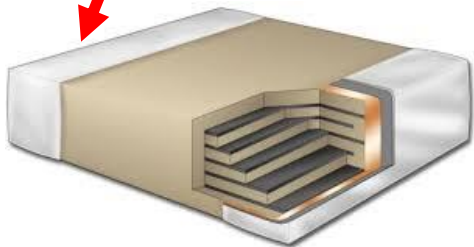
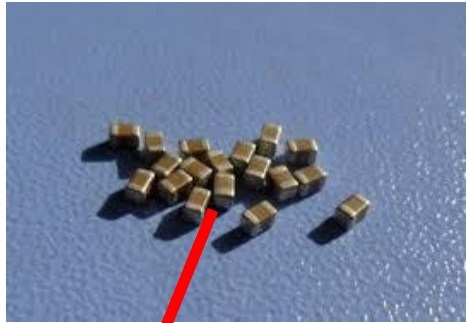
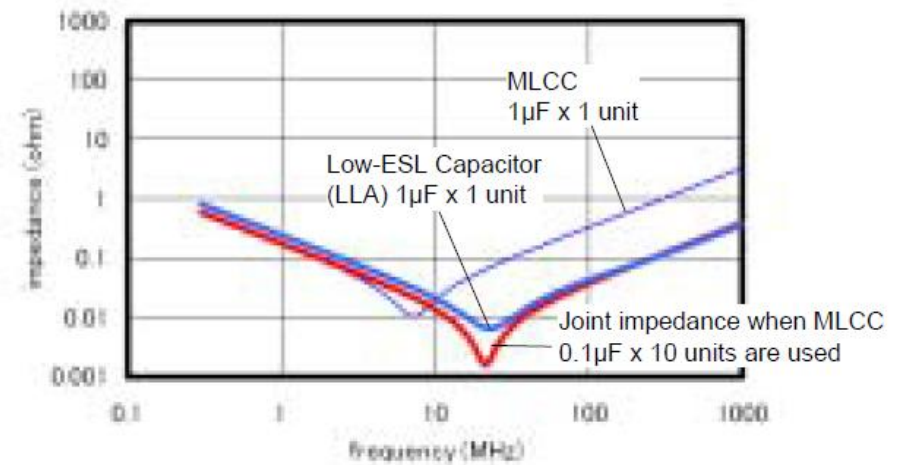
PCB design: decoupling (6)

Capacitors in parallel

small cap. in parallel
better than big one



OR
using low ESL capacitors



(a) LW reverse type



(b) 8-terminal type



(c) 10-terminal type



Reference:
Ordinary MLCC

MLCC: multi-layer-ceramic-capacitor